

Hereditary Cardiovascular Dysplasia*

A Form of Familial Cardiomyopathy

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A SMALL proportion of patients dying suddenly, or following an episode of heart failure, is found at autopsy to show hypertrophy and fibrosis of the myocardium without evidence of significant valvular, hypertensive or coronary artery disease. In some cases it is assumed that the hypertrophy is due to unrecognized hypertension and that the blood pressure has fallen to within normal limits in the terminal stage of heart failure. In others, an excessive alcohol intake is considered to be responsible for the cardiomegaly. In still others, the hypertrophied and fibrotic myocardium is only part of a generalized disease, such as Friedreich's ataxia [1,2] or progressive muscular dystrophy [3-6], and the diagnosis is readily made on clinical grounds. There remains a small group of cases of undetermined origin in which the manifestations of disease are confined to the heart. These cases are labelled as idiopathic cardiac hypertrophy, or heart disease of unknown cause, and their pathogenesis remains obscure and the subject of much speculation [7-11].

In recent years, idiopathic cardiac hypertrophy has been reported to show a familial incidence which may extend over several generations [9,13-30,34]. Our interest in this condition began in the Fall of 1957 following the coincidental admission to hospital of two brothers, aged thirty-nine and forty-one years, (Cases 11 and 12) with unexplained cardiomegaly. In addition to their large hearts these two brothers showed evidence of cerebral damage which was considered to be on a vascular basis. An unusual incidence of sudden death and cerebrovascular accidents in other members of the family led us to suppose that this condition

might be familial in nature. With this possibility in mind, we examined seventy-seven members of the family during the following two years. Complete histories were obtained and physical, electrocardiographic and radiologic examinations performed; these were repeated one year later on the majority of those subjects who had shown abnormalities on the initial screening. In addition, all available information was obtained on those members of the family who had already died.

The results of this survey are summarized in the form of a genealogical chart. (Fig. 1.) Of the seventy-seven persons examined, twenty were considered to have heart disease, and fifty-seven showed no abnormality. In addition, histories obtained on the deceased members of the family strongly suggested that another ten had died of heart disease. Thus a total of thirty members of the family are considered affected, and their case reports are summarized at the end of the paper. Cases 1 to 10 inclusive represent those persons who died before the study was initiated. In two of these we were able to obtain electrocardiograms which substantiated the histories of heart disease obtained from relatives. The remaining eight either had symptoms suggesting heart disease or had died suddenly at an early age.

Since the beginning of the study in 1957, five deaths have occurred among members of the family who had been examined, and autopsies were performed on three of these. In none of the ten members (Cases 1 to 10) who are believed to have died of the disease before this investigation began was a postmortem examination made.

Genealogy. A ninety-five year old member of the family supplied us with much of the early

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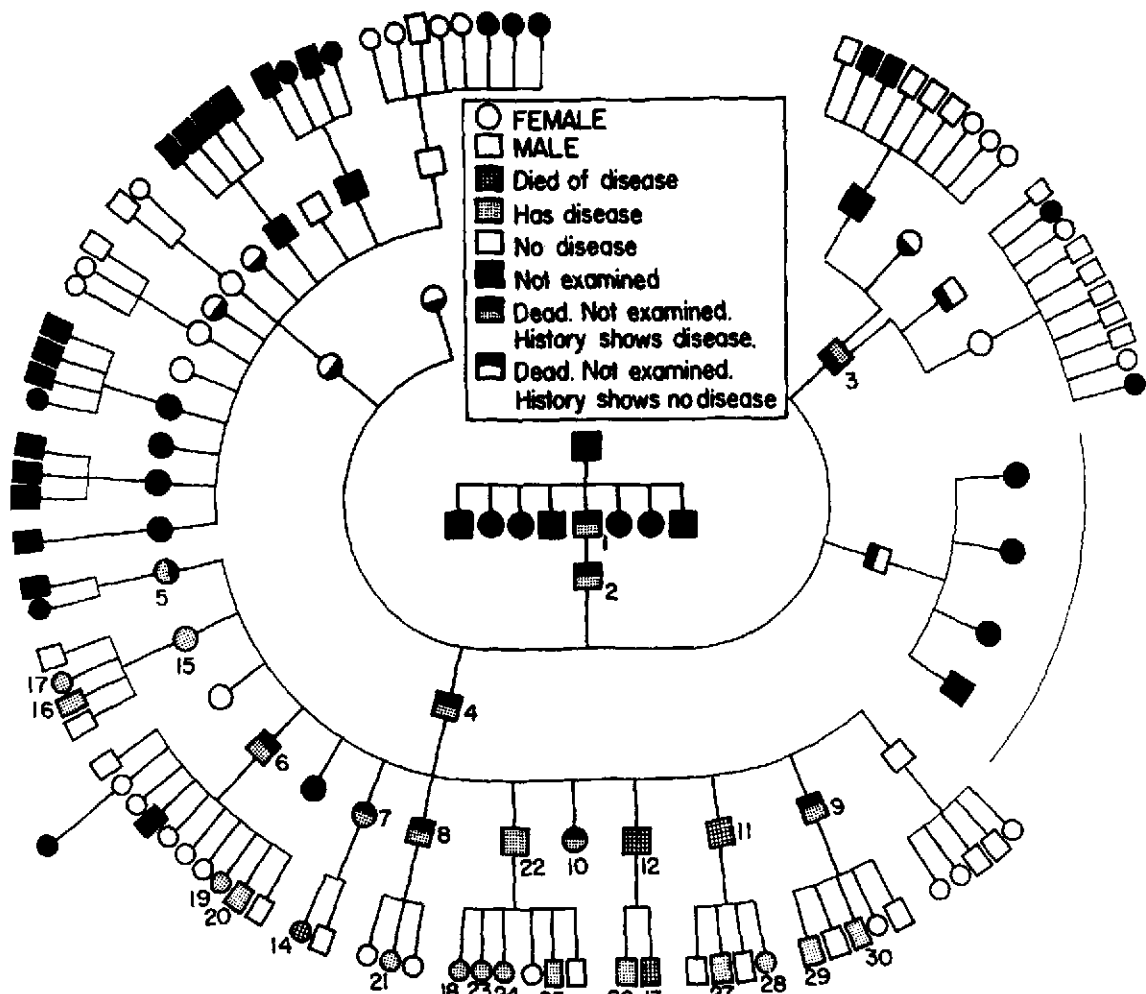


FIG. 1. Genealogical chart.

history, including that of his uncle's sudden death in 1860 (Case 1), and a description of his cousin's heart disease and early death (Case 2). This aged man, who had an excellent memory, was not aware of any other sudden deaths or heart disease in his uncles or aunts, and remembered them as having lived beyond the fiftieth year with the sole exception of the uncle (Case 1) whom we believe to have had the first case of heart disease in this family. (Fig. 1.) The son of this man (Case 2), an only child, had five offspring, and sudden death and heart disease were described as having occurred in two of his sons (Cases 3 and 4). One (Case 3) died suddenly while working on his farm; he had married twice and had twenty-three descendants. We examined sixteen members of this particular branch of the family without obtaining any

evidence of heart disease. A sister of this man died at the age of sixty-four years in a mental institution and was not known to have had heart disease. A second sister died at the age of seventy and showed auricular fibrillation and peripheral embolism at the time of her death; fifteen of her forty-two descendants have been examined and are considered to have normal hearts. A brother emigrated to the United States, where he died at the age of seventy from a cerebrovascular accident. We have not examined this man's progeny as yet but have obtained information through correspondence which has led us to believe that they enjoy good health. The other brother (Case 4), who was disabled from a heart condition and died at the age of forty-six years, is the ancestor of the remaining twenty-six persons who are considered to have cardiac disease.

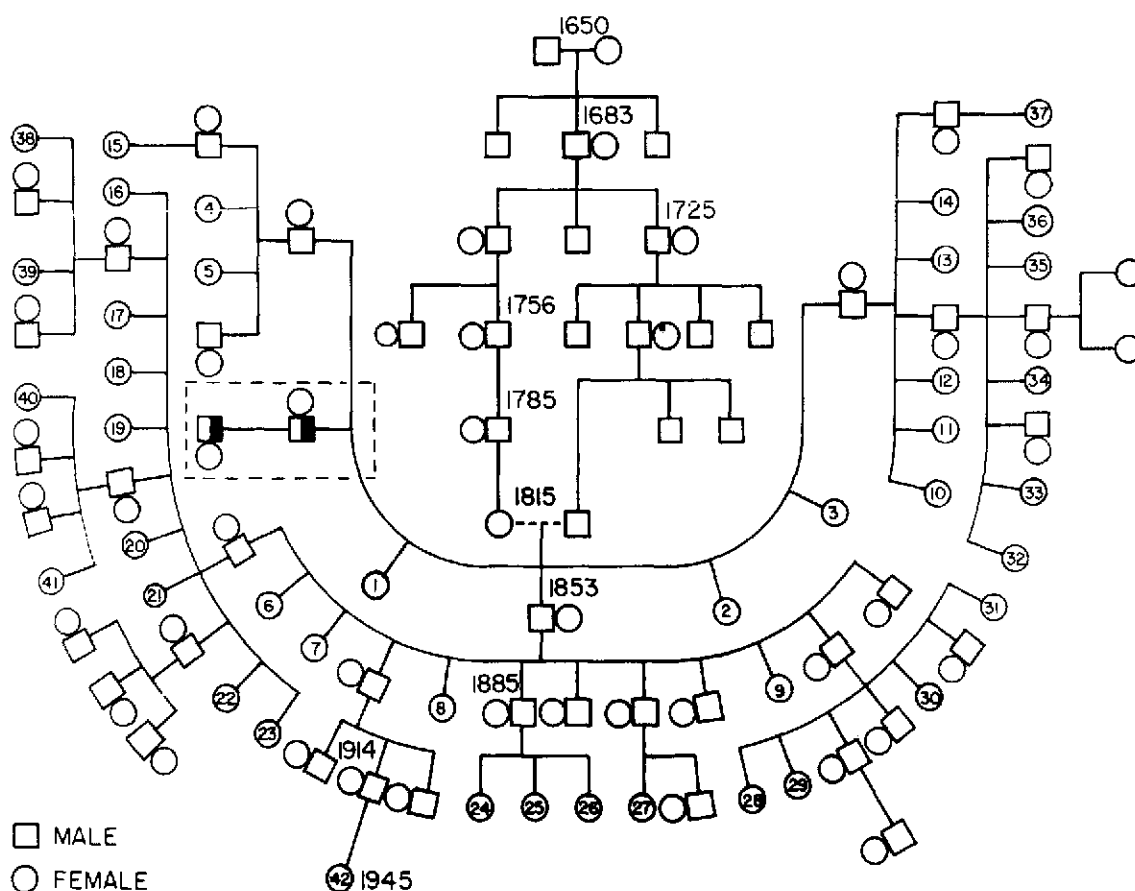


FIG. 2. Genealogical chart. A marriage date is shown in each generation. The rectangular area within the broken lines contains Cases 1 and 2 whose complete line of descent is shown in Figure 1. The numbers in the circles refer to a separate list of married names for follow-up purposes.

Forty-six members of this branch of the family have been examined; twenty have been found to have cardiac abnormalities leading to death in five instances.

While in the process of investigating this family tree we were presented with a genealogical survey of French Canadian marriages in the lower St. Lawrence region [12]. With this additional information we were able to draw up a more complete family tree which dated back to the original immigrant from France who was married in the Province of Quebec in 1650. (Fig. 2.) It will be seen that the original man with heart disease was the offspring of a marriage of two cousins once removed, and that three brothers and three sisters were also married and had numerous descendants. From this survey we obtained the married names of the female members of this family. This information may prove useful in the future to determine the

relationship to our own family of other family groups showing heart disease.

Symptoms. The main symptoms and physical signs found in the affected members of the family are shown in Table I. The available data on those who already had died of heart disease are recorded in Table II.

Shortness of breath was the most common complaint. In most patients it was first noted in early childhood (Cases 13, 14, 17 and 21), but in those living to the fourth decade it did not usually appear until a few months before death (Cases 4, 6, 8, 9, 11 and 12). Weakness was another common symptom but its association with dyspnoea made its evaluation difficult. Transitory episodes of confusion, weakness and giddiness were complained of and were often described as a sensation of "blackness before the eyes." In seven cases this symptom was associated with syncopal attacks, usually precipitated

TABLE I
SUMMARY OF DATA ON THE PATIENTS STUDIED

Case No.	Sex	Age (1958) (yr.)	Symptoms					Signs				Electrocardiogram				Roentgeograms							Cholesterol (mg. %)	Notes			
			Dyspnoea	Weakness	Syncope	Palpitation	Cough	Systolic Thrill	Double Apex Impulse	Atrial Sound	Third Sound	Systolic Murmur	Diastolic Murmur	Abnormal P Wave	High QRS Voltage	Intraventricular Block	Abnormal Q Wave	Abnormal S-T or T (Lone)	Right Atrium	Right Ventricle	Left Atrium	Left Ventricle			Small Aorta	A-V Notch	Pulmonary Congestion
11	M	39	+	+		+	+											2	2	2	3	+	+	+	110	CVA, DA	
12	M	41	+	+														±	±	±	±	+	+	+	104	CVA, DA	
13	M	17	+	+	+	+						3	+		+	+	+	±	±	1	1	+	+	+	112	B.P. 114/90, D	
14	F	19	+	+		+	+				+	3						±	±	1	2	+	+	+		DA	
15	F	50	+	+		+	+										+										
16	M	17										1				+											
17	F	18	+	+	+	+	+					1	+	AF				2	2	2	2		+	+		POS, D	
18	F	17										1				+											
19	F	11								+	+						+	1	1	1	1						
20	M	10	+	+								3					+	1	1	1	1	±	+	+			
21	F	11	+	+						+	+	2	+				+	1	1	1	1	+	+	+	112		
22	M	43							+	+		2					+	1	1	1	1	+	+	+	212		
23	F	16	+								+	1				+		1	1	1	1		+	+	171		
24	F	14								+	+	2					+	±	±	1	1	±		+	+	204	
25	M	10	+	+					+	+	+	2						1	±	±	1	±		+	+	141	
26	M	15	+					+	+	+		4	+	+	+		+	1	1	1	1	+	+	+	153		
27	M	13	+		+						+	1					+	1	1	±	1		+	+			
28	F	18	+	+			+		+		+	2					+	1	1	±	1		+	+			
29	M	15	+	+	+	+	+		+	+	+	3	+	+	+		+	1	1	1	1	+	+	+	96	B.P. 120/80	
30	M	12	+					+	+	+	+	4	+		+	+		1	1	1	1	+	+	+	106		

NOTE: Systolic murmurs graded 1-6; chamber enlargement (x-ray) graded 1-3. D = Died. A = Autopsy. AF = Atrial fibrillation. OS = Opening snap. CVA = Cerebral vascular accident.

TABLE II
AVAILABLE DATA ON MEMBERS OF THE FAMILY WHO DIED PREMATURELY AND PRESUMABLY OF
HEART DISEASE BEFORE OUR STUDY COMMENCED

Case No.	Age at Death (yr.)	Dyspnoea	Cerebral Vascular Accident	Sudden Death	Comments
1	30			+	
2	36	+		+	
3	37			+	
4	46	+	+		
5	45	+	+		Dropsy
6	40	+	+		Congestive failure, syncope, aortic diastolic murmur, A-V block and wide QRS
7	33	+		+	
8	43	+	+		Chest pain and haemoptysis, wide P waves, right bundle branch block
9	35	+	+	+	Blackouts
10	12	+		+	

by exertion. Three patients, aged thirty-nine, forty-one and forty-three, complained of chest pain, which was regarded as due to myocardial ischaemia. Gastrointestinal symptoms consisting of prolonged attacks of anorexia, nausea and

vomiting were noted in five instances. Evidence of congestive failure was present in eight of the fifteen persons who died; eight died suddenly and did not display signs of cardiac decompensation preceding their deaths.

Physical Findings. On physical examination these people were found to be normally developed and of normal intelligence. This was as true of those who were restricted in their activity as it was of those who were asymptomatic. With the exception of slight scoliosis of the thoracic spine in two cases, polythelia in one, and the finding of chronic eczema in one branch of the family, the abnormalities were confined to the cardiovascular system.

A regular rhythm with a normal pulse rate was present in every case but one, an eighteen year old girl (Case 17) who had atrial fibrillation. Two others had occasional ventricular extrasystoles. The blood pressures were within normal limits with one exception (Case 13), in which there was a low diastolic pressure (110/40 mm. Hg).

In five patients cardiac enlargement was detected clinically. In four, a bifid apical impulse could be felt, the first lift occurring with a pre-systolic sound and the second with the usual first sound. Two patients had apical systolic thrills.

On auscultation, the first and second heart sounds were judged to be within the normal range. Nine subjects had an atrial sound at the apex, or at the lower left sternal border. This particular finding was noted in two of the five subjects who have died since our first examinations. A third heart sound was commonly heard but since most subjects were within an age group when this finding may be considered physiologic, its significance is doubtful.

Fifteen of the twenty patients examined had systolic murmurs which were usually widespread but of greatest intensity at the apex or in the lower sternal area. Clinically these were thought to be of the ejection type; this was confirmed by phonocardiography in three cases. (Fig. 3.) Two men (Cases 6 and 12) had early diastolic murmurs. In six instances, in addition to mid-systolic murmurs, short, low-pitched, mid-diastolic murmurs were heard at the apex. These were clearly heard and were distinctly longer than a third heart sound.

Those subjects who were judged to have advanced disease on the basis of their symptoms, electrocardiographic and radiologic findings, usually showed most of these abnormal sounds and murmurs. (Fig. 3.)

Electrocardiographic Findings. Electrocardiograms (Fig. 4) were abnormal in all but one of the twenty members considered to have heart disease. In two instances the diagnosis was

based on this finding alone (Cases 16 and 18). There was close correlation between strong clinical evidence for disease and gross changes in the electrocardiographic tracings.

Repeat electrocardiograms were obtained in thirteen patients one year following the initial examinations. In seven no significant changes were noted. In the remaining six the T waves in leads aVF and V_4 to V_7 had become flattened or inverted.

P waves: Atrial fibrillation was present in one patient (Case 17); in five others (Cases 11, 12, 13, 26 and 29) P waves were judged to be abnormal on the basis of high voltage and notching.

A-V conduction: The P-R interval was normal in all patients. Complete heart block appeared during the last few weeks of life in the patient who was fibrillating.

QRS complexes: QRS complexes were abnormal in nine tracings. Four showed complete bundle branch block, three left, and one right. In several patients an abnormal degree of QRS notching was present without widening of the complex.

In three, there was high voltage of the QRS complexes suggesting left ventricular hypertrophy. Four patients had deep and abnormally wide Q waves.

S-T segments and T waves: The S-T segments were depressed and sagging in the left chest leads in two patients, neither of whom was receiving digitalis.

The T waves were always abnormal. In most cases, flat or diphasic T waves were the only abnormalities recorded. (Fig. 4.)

Radiologic Considerations. All the members of the family were examined both fluoroscopically and radiographically. If the heart was judged to be abnormal on fluoroscopy, roentgenograms of the chest were obtained in posteroanterior, right and left anterior oblique, and lateral projections, within the oesophagus outlined by barium sulphate; those patients in whom the heart was regarded as normal only posteroanterior roentgenograms were obtained.

No significant abnormalities of the diaphragm or pulmonary parenchyma were observed, nor was there evidence of pleural effusion. In three patients (Cases 11, 12 and 17) there was pulmonary venous engorgement and mild pulmonary oedema as indicated by basal septal lines and by increased size and loss of definition of the pulmonary vasculature. The calibre of hilar and peripheral pulmonary arteries, how-

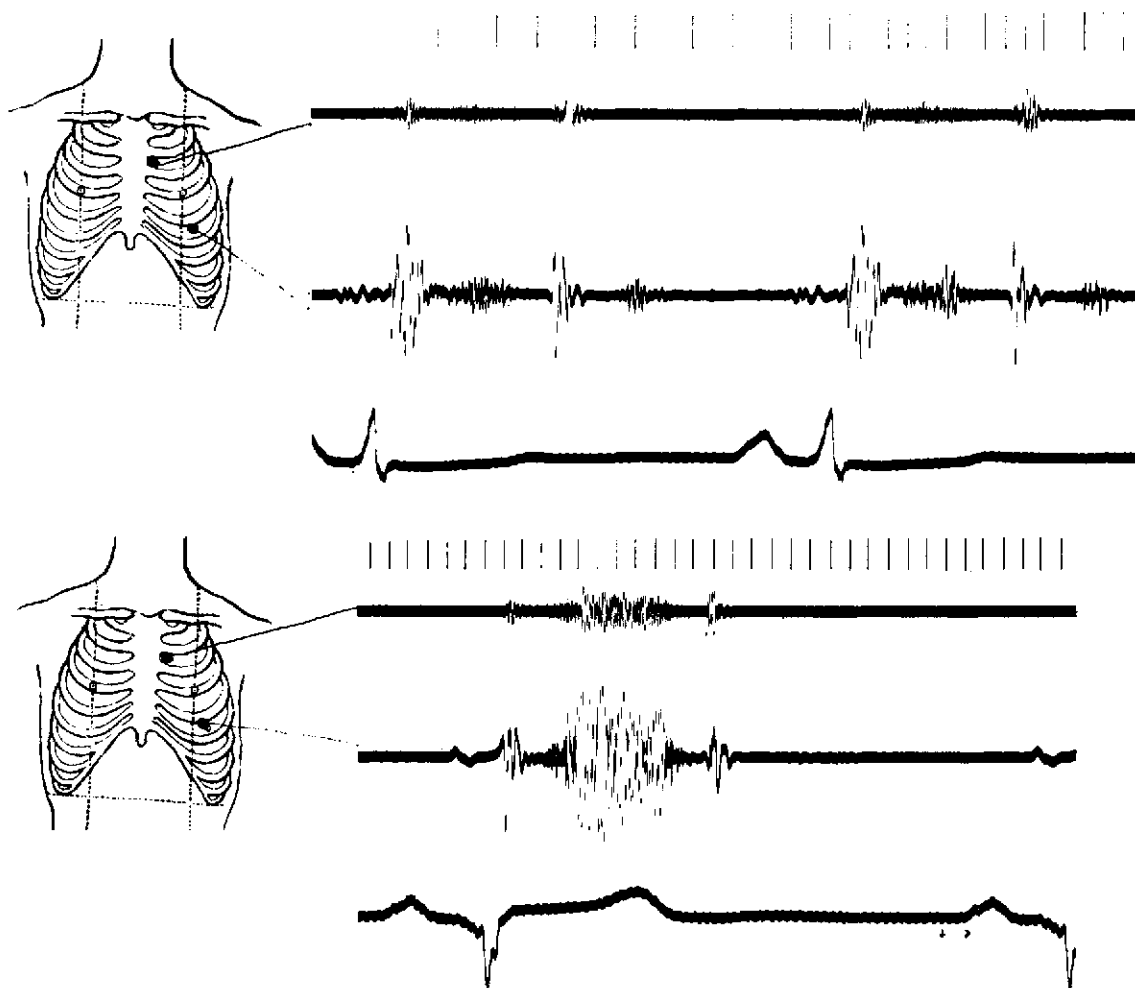


FIG. 3. Phonocardiograms. *Top*, Case 26. Note presystolic sound, ejection systolic murmur, mid-diastolic murmur. *Bottom*, Case 29. Note presystolic sound, ejection systolic murmur and mid-diastolic murmur.

ever, did not suggest the presence of pulmonary arterial hypertension.

Although the degree of cardiac enlargement varied to some extent in affected patients, alteration in contour showed considerable uniformity. In the majority, all chambers were judged to be enlarged (Figs. 5 and 6), although predominant enlargement was consistently left ventricular and left atrial. Although the apex of the heart was displaced slightly to the left and downwards, the main evidence for left ventricular hypertrophy was an outward bulging of its posterior surface, seen to best advantage in the left anterior oblique projection. When this left ventricular bulging was combined with the increased prominence of the left atrial shadow in this obliquity, the position of the atrioventricular

junction was readily identifiable as an indentation or notch. (Figs. 5C, 6C and 7.) This notch was so distinctive as to be clearly visible in all but two of those with cardiomegaly (Cases 19 and 25); in these two only minimal cardiac enlargement was present.

Increased prominence of the anterior surface of the heart in the right anterior oblique projection was present in every patient with cardiomegaly and was interpreted as representing right ventricular hypertrophy.

The aorta (Figs. 5 and 6) was regarded as being of reduced calibre in eight patients. The amplitude of cardiac pulsation was normal, with the exception of a girl with atrial fibrillation (Case 17).

In twelve subjects an opportunity was af-

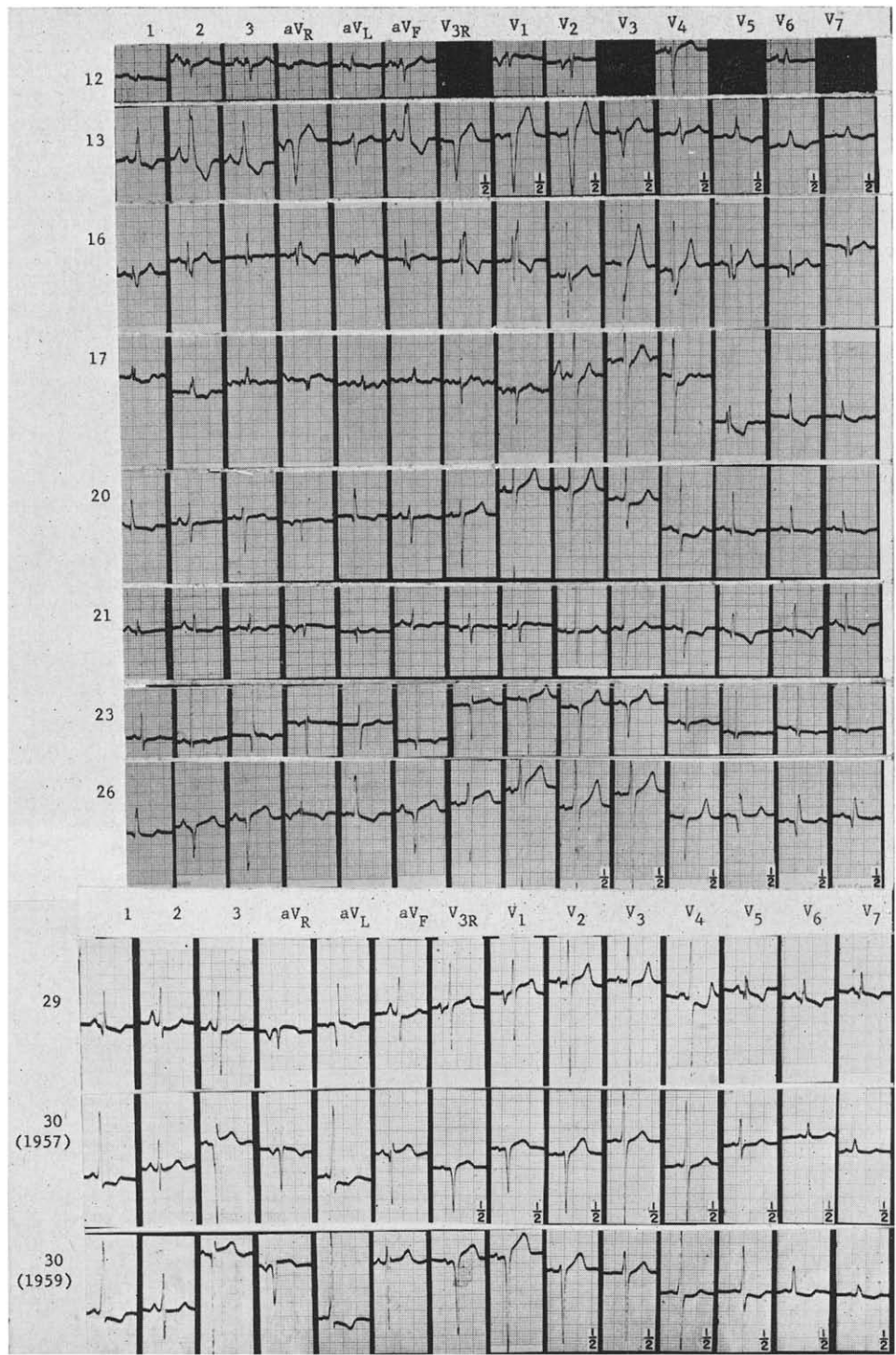


FIG. 4. Electrocardiograms.

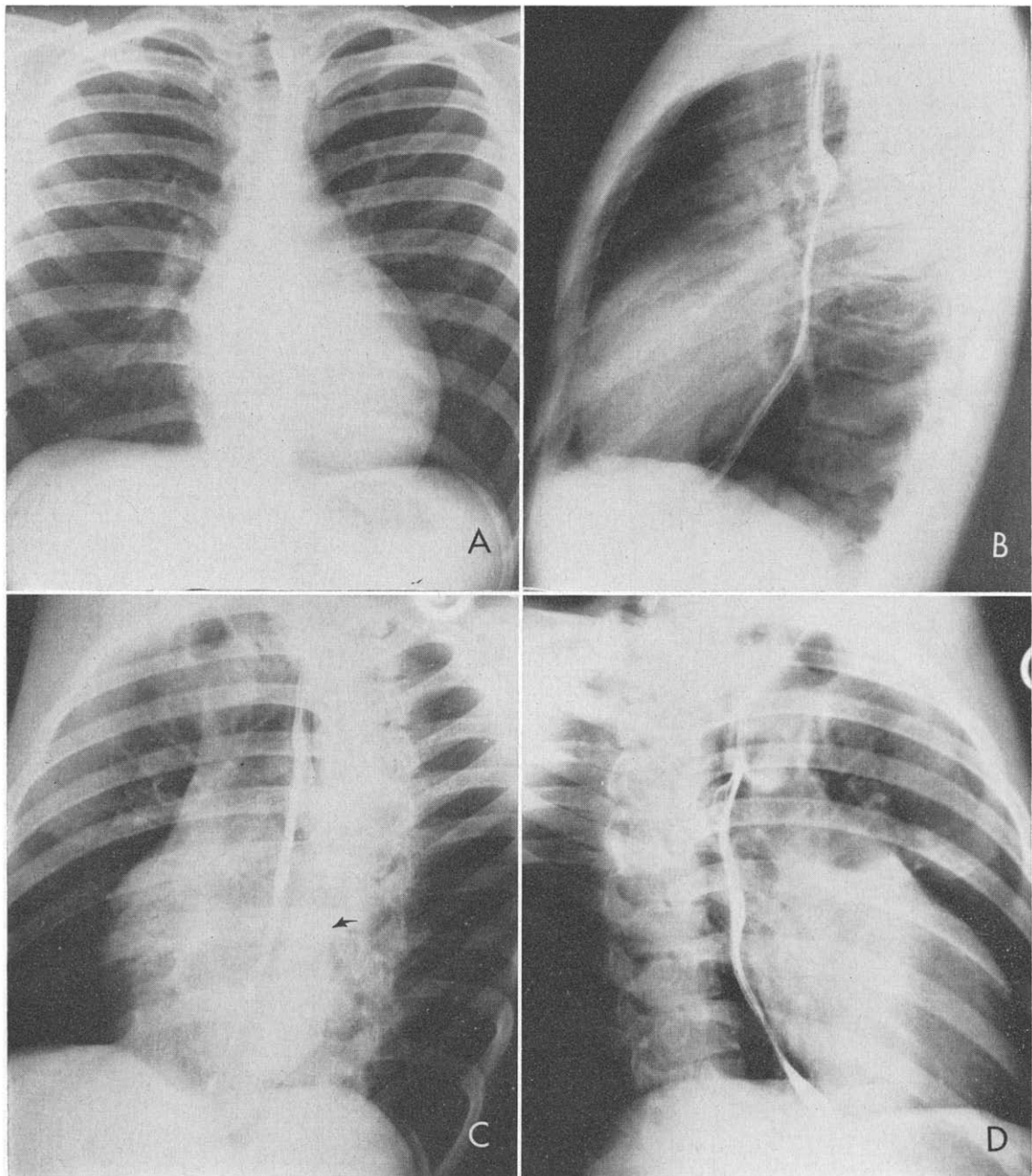


FIG. 5. Case 30. (A) posteroanterior, (B) lateral, (C) left anterior oblique and (D) right anterior oblique views of the chest. Note enlargement of all cardiac chambers, indentation on posterior surface of heart in (C) (arrow), representing left atrioventricular junction.

forded to re-examine the heart at one year's interval, and in four of these (Cases 21, 26, 28 and 30) definite enlargement had occurred during this period. The remainder showed no change.

In addition to fluoroscopic and roentgeno-

graphic examinations of the heart, tomograms of the lungs and roentgenograms of the skull were obtained in three young boys (Cases 26, 29 and 30). These procedures were carried out as part of a thorough examination to assess the pulmonary arterial calibre and to detect the

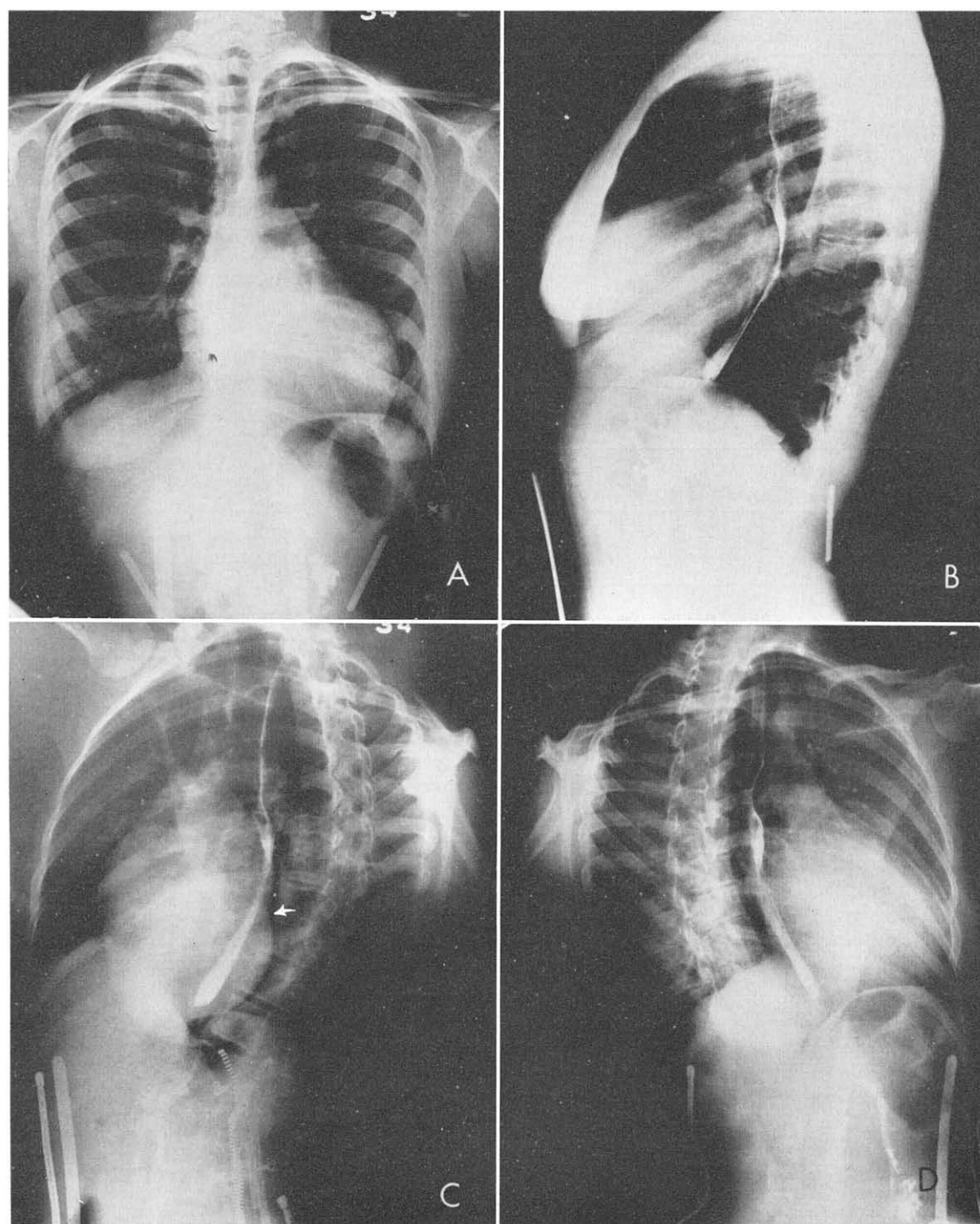


FIG. 6. Case 14. (A) posteroanterior, (B) lateral, (C) left anterior oblique and (D) right anterior oblique views of the chest. All cardiac chambers are enlarged. Note particularly indentation on posterior surface of heart in (C) (arrow), representing left atrioventricular junction. Compare with Figure 7.

TABLE III
MEASUREMENTS OF HEART (NORMAL VALUES INSERTED
IN BRACKETS)

Area	Thickness (in mm.)			
		Case 11	Case 13	Case 14
Left ventricle	At the base	13 (8-10)	12 (8-10)	22 (8-10)
	Mid-portion	13-14 (10-12)	20 (10-21)	20 (10-12)
Right ventricle	At the base	5 (2-3)	7 (2-3)	5 (2-3)
	Mid-portion	4 (3-4)	8 (3-4)	5 (3-4)
Interventricular septum	At the base	11 (10-12)	32 (10-12)	30 (10-12)
	Mid-portion	10-11 (12-14)	26 (12-14)	26 (12-14)

presence of associated abnormalities. The roentgenograms of the skull and the central and peripheral pulmonary vessels were judged to be within normal limits.

Special Investigations. Five members of this family have been admitted to the Royal Victoria Hospital. In 1957 the two original patients (Cases 11 and 12) underwent a variety of laboratory procedures and the results of these tests were within normal limits, with the exception of the serum cholesterol levels which were low. (Table I.) In 1959, three younger boys (Cases 26, 29 and 30) were admitted for more complete investigation, and the following examinations were carried out: phonocardiograms; cardiac output by a dye method; brachial artery pulse tracings; pulmonary function tests; electroencephalograms; tuberculin, histoplasmin and toxoplasmin skin tests; blood volumes; blood cytology, including blood grouping; blood coagulograms; protein electrophoresis; serum electrolytes; serum cholesterol levels; blood sugars; blood Wassermann tests; urinalyses; twenty-four-hour amino acid content of urine.

Abnormal results included a lowered cardiac output in one patient (Case 29); slightly reduced vital and functional residual capacities in two (Cases 29 and 30); a paroxysmal dysrhythmia seen on the electroencephalogram of one (Case

29), and low serum cholesterol levels. Serum cholesterol levels were subsequently determined in six other members of the family, and were found to be low in two. Thus of the eleven in whom serum cholesterol levels were determined, six showed values well below the normal range. (Table I.) The brachial artery tracings were regarded as abnormal and are described subsequently and compared to those reported in the literature.

Pathologic Findings. Autopsies have been performed on three members of the family. A seventeen year old boy (Case 13) and a nineteen year old girl (Case 14) died suddenly without having shown any evidence of cardiac failure during life. A man of thirty-nine years (Case 11) died following recurrent cerebrovascular episodes which had rendered him aphasic and hemiplegic.

The autopsies on the two young people were performed after their bodies had been embalmed, with resultant alteration in the appearance of some organs. The pathologic findings in these two instances were remarkably similar and were confined to the cardiovascular system. The boy's heart weighed 575 gm., the girl's 505 gm. All chambers were hypertrophied (Table III), particularly the left ventricle, and in each case there was a peculiar nodular muscle mass in the cephalic two-thirds of the interventricular septum. (Figs. 9 and 10.) This hypertrophied myocardial mass bulged into both ventricular cavities and merged gradually with the more symmetrically hypertrophied muscle fibres in the apical portion of the septum. (Figs. 7, 8 and 11.) The papillary muscles and trabeculae carneae cordis were enlarged and elongated. The valves were normal and the endocardium smooth and transparent. Small and large irregular patchy areas of fibrosis were seen throughout the myocardium, especially in the left ventricle, papillary muscles and interventricular septum. (Figs. 9 and 10.) The coronary arteries showed no evidence of occlusion and were normal in distribution and calibre.

The microscopic findings in two patients (Cases 13 and 14) were identical. A marked hypertrophy of muscle fibres was noted. In addition, many of these fibres showed disintegration of cytoplasmic components and loss of striations; in some areas, only shadow structures of the sarcolemma remained. This process of transition from hypertrophied to disintegrated

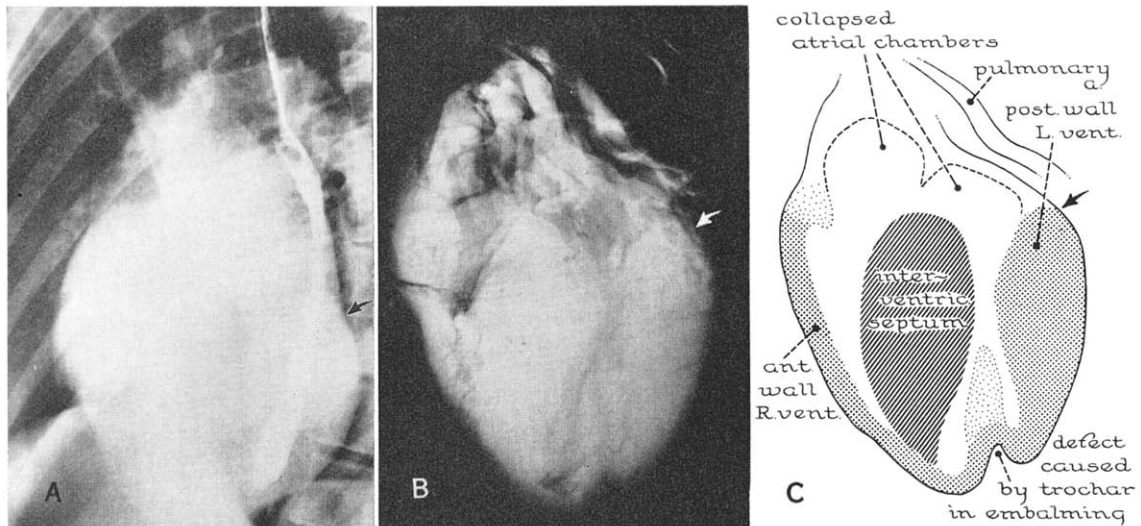


FIG. 7. Case 14. (A) left anterior oblique roentgenogram, (B) roentgenogram of gross specimen of heart in approximately same projection as in (A), (both atria are collapsed); (C) diagram of (B). Note generalized hypertrophy of left ventricle, localized muscle mass in cephalic portion of interventricular septum bulging into both left and right ventricular chambers and simulating hypertrophy of right ventricle, notch or indentation on posterior surface of heart in (A), corresponding to upper edge of hypertrophied left ventricular muscle mass in (B) (arrows).

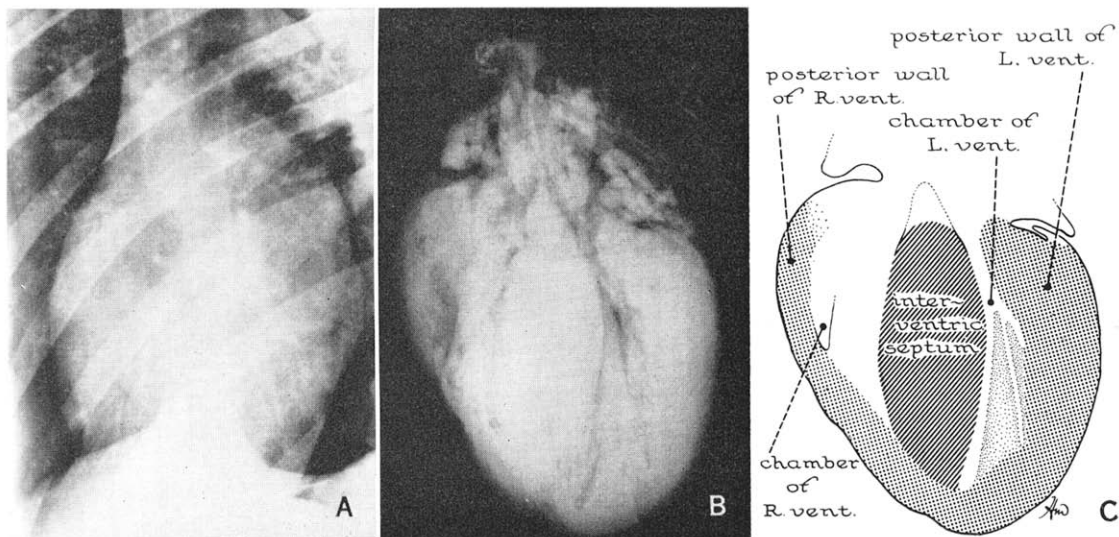


FIG. 8. Case 13. (A) left anterior oblique roentgenogram, (B) roentgenogram of gross specimen of heart in approximately same projection as in (A) (both atria are collapsed), (C) diagram of (B). Generalized hypertrophy of left ventricle is marked but the asymmetrical muscle mass in cephalic portion of interventricular septum is not as striking as in Case 14, Figure 7.

fibre occurred gradually, without any sharp delineation. (Fig. 12.) In the left ventricle, and particularly in the nodular mass of the interventricular septum, there were areas where the muscle fibres formed an interdigitating network of "Y" shaped and semicircular bundles

in a peculiar haphazard architectural pattern. In these areas, loosely arranged connective tissue, which included blood vessels, was seen to lie between these muscle bundles. (Figs. 13 and 14.) Throughout the myocardium there were irregular areas of fibrosis. (Fig. 15.) No



FIG. 9

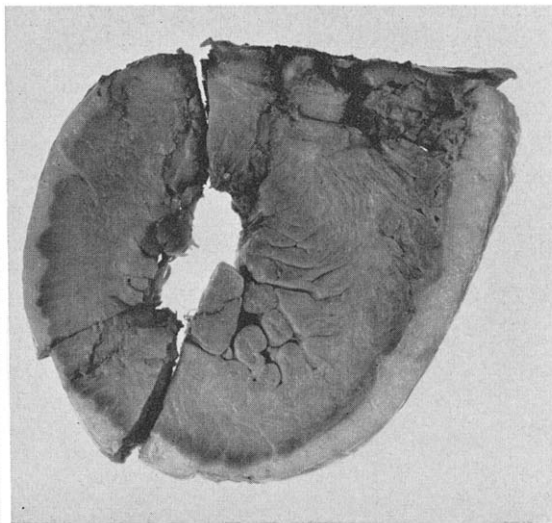


FIG. 10

FIG. 9. Case 13. Coronal section of the heart showing hypertrophy of both ventricles and a muscular mass in the interventricular septum. Multiple areas of fibrosis are scattered throughout the myocardium.

FIG. 10. Case 14. Coronal section of heart showing hypertrophy of the left ventricle with a muscular mass in the interventricular septum. There is distortion of both ventricles caused by the embalming process.

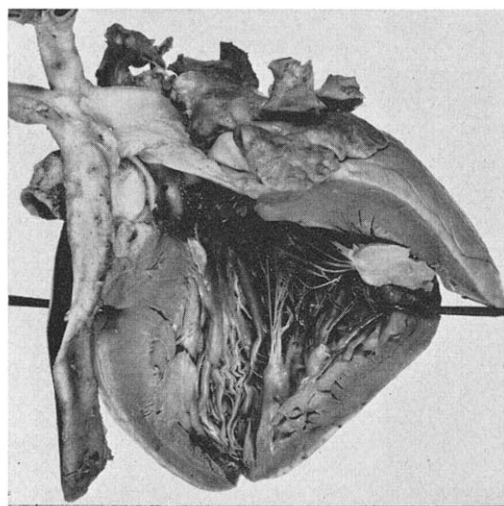


FIG. 11

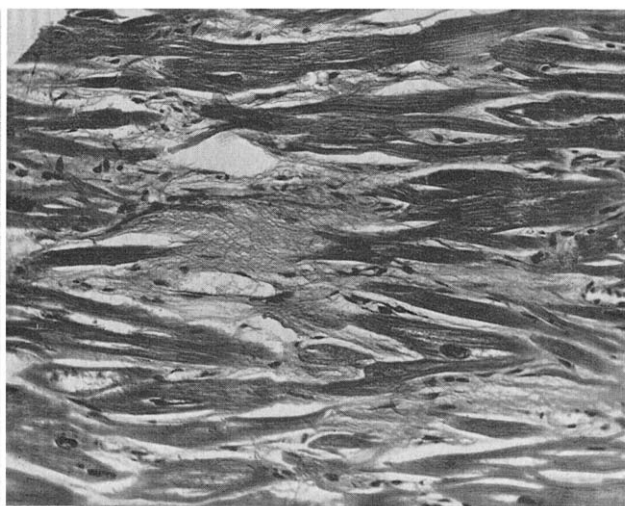


FIG. 12

FIG. 11. Case 13. The heart with an opened left ventricle showing hypertrophy of the myocardium. Muscular mass of the interventricular septum bulges into the ventricle.

FIG. 12. Case 13. Section through the myocardium of the left ventricle. There is disintegration of muscle fibres with replacement by connective tissue. Hematoxylin and eosin stain.

inflammatory reaction of any type was found in multiple sections.

The third autopsy was performed on a thirty-nine year old man (Case 11) whose heart weighed 680 gm., after fixation. (Fig. 16.) There was dilatation of all heart chambers and con-

centric hypertrophy of the left ventricle. (Table III.) The localized hypertrophy of the interventricular septum seen in the two other patients was not observed. Many large areas of fibrosis were scattered throughout the left ventricular myocardium. These fibrotic areas tended to

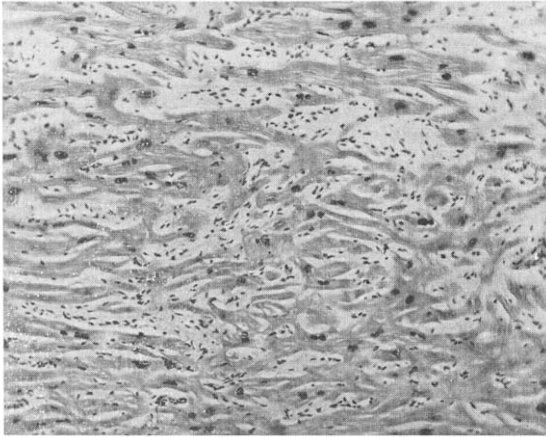


FIG. 13

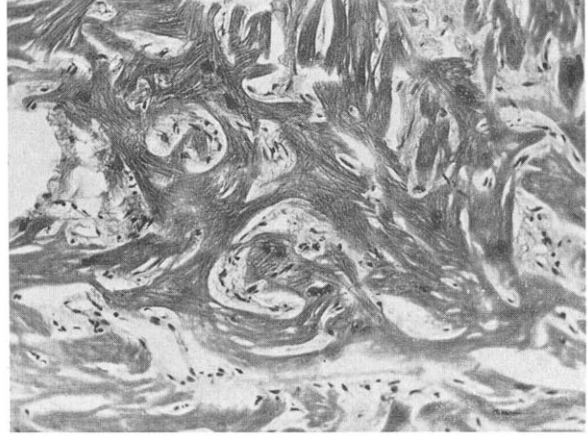


FIG. 14

FIG. 13. Case 14. Section through the myocardium of the left ventricle. Muscle fibres are hypertrophic and separated by loose connective tissue. Hematoxylin and eosin stain.

FIG. 14. Case 13. Section through the myocardium of the left ventricle showing bizarre arrangement of intertwining muscle fibres which are separated by loose connective tissue. Hematoxylin and eosin stain.

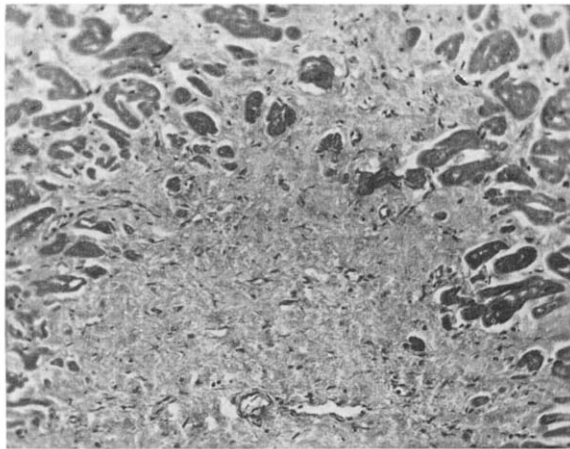


FIG. 15

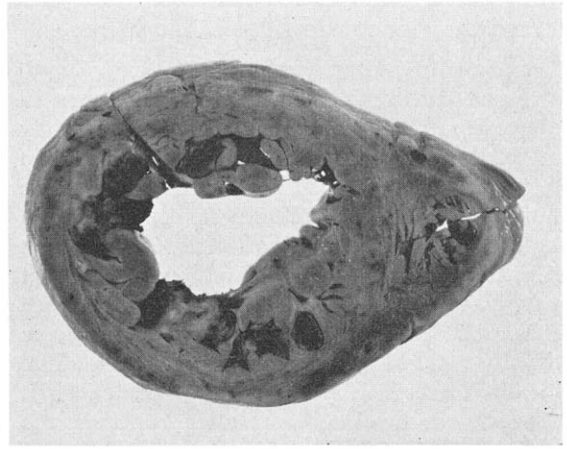


FIG. 16

FIG. 15. Case 13. Section through the myocardium of the left ventricle. There is a large area of fibrosis and hypertrophy of muscle fibres. Hematoxylin and eosin stain.

FIG. 16. Case 11. Coronal section of the heart showing hypertrophy of the myocardium with multiple areas of fibrosis and mural thrombi.

coalesce toward the apex on the anterior aspect of the left ventricle and interventricular septum, where the endocardium was also thickened and fibrosed, with firmly attached mural thrombi. There was slight non-specific thickening of the anterior leaflet of the mitral valve which had not resulted in any stenosis. The other valves were normal in appearance. The coronary arteries were patent throughout although there was a mild degree of arteriosclerosis.

On microscopic examination the myocardial

fibres were hypertrophied and in some places disintegrated and replaced by fibrous tissue. The bizarre architectural pattern seen in the other two patients was not present. Organized and organizing mural thrombi were attached to the area of thickened endocardium at the apex.

The lungs, kidneys, brain and intestines showed evidence of old and recent infarcts.

All three subjects showed hypoplasia of the aorta (Table IV) and medium-sized arteries, with slight intimal arteriosclerosis. There was

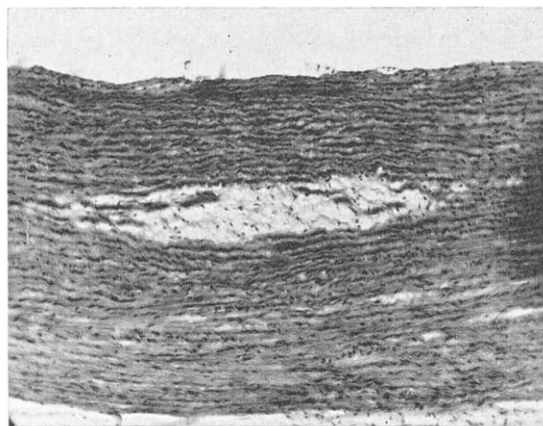


FIG. 17. Case 14. Section of the abdominal aorta. There is focal accumulation of slightly basophilic mucoid substance in the media. Hematoxylin and eosin stain.

atrophy and degeneration of the muscle coats with increase in collagen and accumulation of basophilic mucoid substances. (Fig. 17.)

COMMENTS ON HEREDITARY CARDIOVASCULAR DYSPLASIA

Heart disease has been found to occur in approximately 50 per cent of the children of any afflicted member of this family, independent of sex. Our original suspicion that such a large number of cases in one family might represent a hereditary form of cardiac disease is borne out by this non-sex-linked Mendelian dominant pattern of transmission. The pathologic evidence is consistent with a defect in the embryologic development of the cardiovascular system. The localized massive hypertrophy of the cephalic portion of the interventricular septum, the bizarre disorganized architecture of the myocardial fibres, and the hypoplasia of the aorta and large arteries are not characteristic of any acquired heart disease.

In recent years the medical literature has contained an ever increasing number of descriptions of familial and hereditary heart disease, some of which bear a resemblance to our own cases. The first reference to myocardial disease in a family was made in 1946 by Addarii et al. [73], who described bundle branch block and cardiac failure in two brothers whose mother had died from heart disease at the age of thirty-four years. Postmortem examination of one of the brothers revealed hypertrophy and diffuse fibrosis of the myocardium and hypoplasia of the aorta. In 1949 William Evans [74] described a group of ten patients with idiopathic cardiac

TABLE IV
MEASUREMENTS OF AORTA (NORMAL VALUES* INSERTED IN PARENTHESES)

Area	Circumference (in mm.)		
	Case 11	Case 13	Case 14
Above valve	52 (73)	40 (57)	40 (55)
Underneath arch	45 (54)	35 (44)	30 (43)
At diaphragm	45 (53)	23 (40)	34 (39)
Above bifurcation	26 (41)	20 (31)	(29)

* After KANI, J. *Tabulae Biologicae*, vol. 3. Berlin, 1926. Junk.

hypertrophy. He introduced the term "familial cardiomegaly," since five of his patients were members of two separate families. During the last decade many other reports of familial heart disease have appeared. In some instances the evidence has been based on clinical examination only [9,21,23-27], but in most cases autopsies have been performed and have been found to show myocardial hypertrophy and fibrosis [15-20,22,28-30,34]. Although there is considerable variation in the clinical picture in these reported series, two features are common to all: A familial occurrence which in some cases can be traced over several generations and, in those cases in which autopsy was performed, the invariable finding of hypertrophy and fibrosis of the myocardium.

In the series previously reported the number of cases involved has not been sufficient to establish a genetic pattern. Brigden [27], in reviewing the non-coronary cardiomyopathies, has stated that the published reports of familial cardiomegaly suggest a Mendelian dominant mode of transmission. More recently Brent et al. [30] have described a familial subaortic stenosis in which the family pedigrees indicate a dominant inheritance.

Familial myocardial disease is usually described as occurring in the second or third decades, and the majority of our own patients fell into these age groups. We did not obtain a history of frequent infant death to suggest that a more severe degree of developmental defect might exist, although it is possible that the dis-

ease did not occur in the descendants of our patient (Case 3) due to the deaths in early infancy of two of his offspring. In our own cases it has been more the rule than the exception to live to a marriageable age, and this has resulted, over three generations, in a sixteenfold increase in the number of cases. Similarly, there does not appear to be any clear-cut evidence of the disease occurring at an earlier age in succeeding generations (phenomenon of anticipation) or of it being manifested in more severe forms (phenomenon of potentiation). Most of the children in our series are not seriously disabled and it would seem likely that they will live as long as their afflicted ancestors.

In at least some instances familial myocardial disease appears to be relatively benign and there have been descriptions of older people in the sixth or seventh decade showing little in the way of disability. Soulie and his associates [22] include in their report a fifty-four year old man, followed up over a two year period, who had a transitory episode of right hemiplegia and aphasia and subsequently died suddenly. Facquet et al. [24] report a mild form of familial cardiomegaly and describe a woman showing a minor degree of heart failure at the age of fifty-seven, whose eight children were affected, all of whom were asymptomatic. She also had a brother, fifty-three years of age, who had electrocardiographic and radiologic evidence of heart disease and who was completely asymptomatic. The second family, described by Campbell and Turner-Warwick [17], included two women of sixty-six and sixty-nine years of age, both of whom were considered to suffer from familial heart disease. In our own series the finding of abnormal electrocardiograms in two adults (Cases 15 and 22) with little or no disability suggests that minor involvement may be quite compatible with a relatively normal life span. The occurrence of this heart disease in adults free or almost free from symptoms raises the possibility that the brothers and sisters of our patient (Case 1) may also have been affected and that their descendants may eventually prove to have the disease.

The clinical picture presented by members of this family did not differ materially from that of rheumatic endocarditis, and several of the children had been considered to have valvular disease when seen individually by cardiologists. Dyspnoea was the most common symptom; the severity and the age of onset of this complaint

varied considerably from case to case. Some members of the family attained adulthood before noticing shortness of breath, while others were disabled from early childhood. Sudden brief episodes of loss of consciousness, usually brought on by physical activity, were common occurrences. Syncopal attacks, sudden death [14,15,17-19,22,30] and cerebrovascular accidents ascribed to embolization from mural thrombi [18,22,28] have occurred commonly in previously reported instances of familial heart disease. In some descriptions these syncopal attacks have been noted to occur in subjects with complete heart block [16,17], but more often they have not been associated with any obvious conduction defect and have been of a transitory nature, frequently coming on following exertion.

Electrocardiography proved to be the most sensitive means of detecting the disease in this group. In some cases the tracings were judged to be abnormal on the basis of changes in the T wave alone. The assumption that this could represent an hereditary myocardial disease in an eleven year old boy (Case 20) seems reasonable. On the other hand such changes in adults (Cases 15 and 22) could be explained by myocardial ischaemia secondary to coronary artery disease, and for this reason the diagnosis of primary myocardial disease in these two subjects is slightly suspect. The more bizarre electrocardiographic patterns were found in those who had clinical evidence of a more severe form of the disease. High amplitude of the QRS complexes, pathologic Q waves, and varying degrees of bundle branch block were commonly seen. In contradistinction to our own findings, the literature on familial heart disease reveals arrhythmias to be a rather common feature. Sinus bradycardia [14,17] and nodal rhythms [15,17] have been described, as well as auricular fibrillation and flutter [14,17,19,22,29]. Varying degrees of partial A-V block [14,17,20-23,40], and a few cases of complete atrioventricular block [14,16,17] and Wolff-Parkinson-White syndrome [22] have been noted. The most characteristic electrocardiographic findings, however, have been bundle branch blocks [14,17,18,22,23,25,28] and a high amplitude of the QRS complexes in the pattern of left ventricular hypertrophy [16,22,24-26,29,34].

Radiologic examination of the heart revealed cardiomegaly in fifteen of the twenty patients. Although the cardiac enlargement appeared to involve all chambers of the heart, the

left ventricle and left atrium were predominantly affected. A very characteristic finding seen in all but two of those patients showing cardiomegaly was an indentation visible in the left anterior oblique projection and considered to represent the atrioventricular junction between hypertrophied left atrium and left ventricle. This indentation has also been noted occasionally in patients with rheumatic mitral valvular disease with combined stenosis and insufficiency, but not as commonly as was observed in this group of patients. The prominence of the anterior surface of the heart in the right anterior oblique projection, found in almost every patient with cardiomegaly, was interpreted as representing right ventricular hypertrophy. The subsequent autopsies, however, revealed only minimal right ventricular hypertrophy, and our erroneous interpretation of enlargement of this chamber was readily explained by the large interventricular septal mass bulging forward and producing a prominent anterior cardiac wall.

The results of the laboratory procedures on the patients whom we admitted to hospital were not particularly revealing. The finding of low serum cholesterol levels in some of the patients was unexpected and it is perhaps of some significance that all patients showing hypocholesterolemia were considered clinically to have evidence of a more severe disease. The values obtained are in the range seen only in hyperthyroidism or in advanced liver disease, and none of these patients showed evidence of either of these two conditions. Although liver function tests were not obtained as an investigative procedure, one of the patients (Case 13) with a serum cholesterol level of 112 mg. per cent, subsequently died; his liver was found to be normal at postmortem examination. Similarly, in previously reported series, little information has been obtained from laboratory procedures. Serum cholesterol levels have been measured occasionally and in some instances have been found to be within the normal range [30], although it is of interest that in two case reports [20,43] serum cholesterol levels were abnormally low.

The localized muscular hypertrophy at the base of the septum in our two autopsy cases caused subaortic stenosis and this provides an explanation for the ejection systolic murmurs heard in the majority of cases. This assumption is supported by the absence of systolic murmurs in the third autopsy case, in which gross septal hypertrophy was not evident. In addition, in

some cases this muscular mass appears to interfere with the inflow of blood from the left atrium, producing a mid-diastolic murmur. The early diastolic murmurs are difficult to explain.

The hypertrophied muscular mass in the left ventricular outflow tract has become a well recognized pathologic entity, and there is reason to believe that this lesion may be acquired [35], although in most series the evidence suggests a congenital origin. Many cases do not show a familial incidence [28,37-36], but it appears to be a very characteristic finding in several families with heart disease. Evans [74] refers to a sixteen year old boy who showed a considerable enlargement "involving the base of the left ventricle" at autopsy. A thirty-eight year old female member of the family described by Davies [15] was found at autopsy to have an area of fibrosis 4 cm. square, located "on the upper part of the interventricular septum, behind the aortic cusp of the mitral valve." Teare [28] has described what he calls "asymmetrical hypertrophy" of the heart in three siblings, two of whom at autopsy showed globular-shaped hearts with hypertrophy of the interventricular septum and the anterior walls of the left ventricles. More recently Bercu et al. [29] have reported on "pseudo-aortic stenosis produced by ventricular hypertrophy." Brent et al. [30] have presented two families with "familial muscular subaortic stenosis," and Braunwald et al. [32] have described familial cases of "idiopathic hypertrophic subaortic stenosis." From the descriptions given in the literature this unusual pathologic finding is probably the same abnormality masquerading under a great many pseudonyms. It has been suggested that this obstructing lesion may be suspected and differentiated from valvular aortic stenosis on clinical grounds [31,32,34]. The ejection murmur is heard at the left sternal border, or at the cardiac apex, and not at the base of the heart. A palpable thrill may be detected over the same area but is not common. The peripheral arterial pulse is normal, but direct pulse tracings show a rapid upstroke time, sometimes followed by a second positive deflection, and not the slow rise to a peak, or the anacrotic notch of aortic stenosis [31,32,34]. Our own findings (including the brachial artery tracings, Figure 18) are in agreement with the reported clinical manifestations of subvalvular muscular stenosis. The rapid initial deflection in the pulse tracing, which is not seen in valvular aortic stenosis, is particularly

significant when a large pressure gradient can be demonstrated between the main cavity of the left ventricle and a systemic artery. Under these circumstances it is likely that symptoms will be present, and Braunwald et al. [32] suggest that operation is indicated. Additional confirmatory evidence that the patient has muscular obstruction rather than valvular stenosis can be obtained by left ventricular angiocardiography [32,34].

Our own patients were not subjected to pressure measurements across the aortic valve or to angiocardiography, since it was our opinion that the slight risk involved in carrying out these procedures was hardly justified in a group of subjects who were prone to sudden death and, as we thought, were not candidates for any remedial surgery. However, in view of the encouraging initial results in one patient reported by Goodwin et al. [34], and two cases reported by Braunwald [32], following an operation suggested by Cleland [34], we are at present making arrangements to carry out these additional diagnostic procedures on some of our family group prior to considering them for operation.

In some reports [13,16-20,22,37] of familial myocardial disease, the cardiac muscle has been found to be symmetrically hypertrophied and no mention is made of a localized mass in the interventricular septum. Our own pathologic findings indicate that even in the same family a diffuse form of myocardial hypertrophy may be found in some cases, and the interventricular septal hypertrophy be predominant in others.

Reference has also been made in the literature to endocardial thickening [19,22] but this pathologic change has been less impressive than the myocardial involvement and hence has received little emphasis. However, the literature also contains descriptions of combined endocardial and myocardial disease with a familial incidence [37-39], in which the former lesion has been predominant; such conditions have been referred to as endocardial fibroelastosis or endomyocardial sclerosis. Brigden [27] has ventured the opinion that this thickening of the endocardium may actually be due to organized thrombi which have become closely interwoven with the adjacent endocardium.

Microscopic examination in our cases showed hypertrophy of individual muscle fibres with scattered areas of fibrosis and degenerating myofibrils, without evidence of deposition of foreign substance or cellular infiltration. These findings

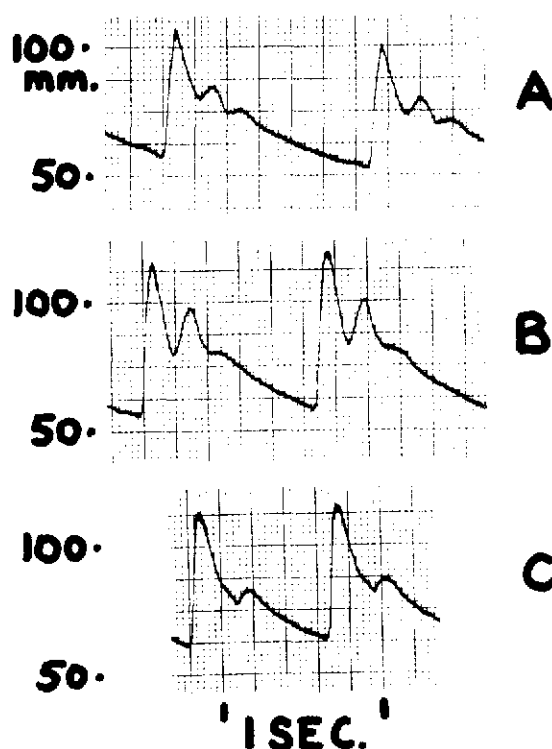


FIG. 18. Brachial pulse tracings. Note the normal upstroke time and the prominent secondary wave. A = Case 26, B = Case 30, C = Case 29.

are in keeping with the majority of published autopsy reports, although Evans [14] and Gaunt and Lecutier [16] found glycogen in the myocardium, and Sommers [19] and Nikkila and Pelkonen [20] have described cellular infiltration. Teare [28] has noted a bizarre arrangement of individual muscle bundles such as was found in two of our patients (Cases 13 and 14).

Hypoplasia of the aorta was detected radiologically in eight members of our family, and measurements made on three of these at autopsy confirmed this observation. (Table IV.) The association of aortic hypoplasia with familial myocardial disease has been previously noted either radiologically [15,24,26] or at post-mortem [15,19,43]. Hayes et al. [43] stressed the finding of a hypoplastic aorta in their family, and described interventricular hypertrophy in a nineteen year old boy. The clinical picture manifested by this patient appears to be identical with that seen by us, even to the finding of a low serum cholesterol level. Microscopic examination of the aortas in our three cases revealed atrophy and degeneration of the media with an accumulation of mucoid basophilic material. A

similar deposition of foreign material is often found in cases of Marfan's syndrome, and it is of interest that this condition may also present as a cardiomyopathy. Whitfield et al. [47] have described a man with arachnodactyly, hypoplasia of the aorta, and myocardial hypertrophy and fibrosis. A familial form of this syndrome, with autopsies showing hypertrophied and fibrotic hearts, has also been reported [40,42].

The disease which we have described is considered to be the result of an inherited defect in development involving the myocardium and the media of larger arteries. Similar if not identical conditions have been reported under a variety of titles, none of which seems adequately to incorporate all significant features of the disease. The terms "familial cardiomegaly" and "familial or hereditary cardiomyopathy" do not take into account the fact that some patients fail to show cardiac enlargement or that the larger vessels may also be involved. In most other reported series of familial myocardial disease, interest has been focused on the localized interventricular hypertrophy with its resultant subaortic stenosis, and a great number of descriptive terms have been suggested for this lesion. In view of the lack of a suitable name for this well delineated clinical and pathologic entity, and recognizing the desirability of incorporating in the title some reference to the involvement of the larger blood vessels, we propose that this disease be known as "hereditary cardiovascular dysplasia."

SUMMARY

A large family is described in which a high incidence of heart disease has been found (idiopathic cardiomegaly). Of a total of eighty-seven members so far considered (seventy-seven of whom have been examined, together with ten who are presumed to have died of the disease before the study commenced), thirty have been found to be affected.

A genealogic survey dating back to the arrival of the original immigrant from France in the early seventeenth century has allowed certain conclusions to be drawn with regard to hereditary characteristics. The pattern of occurrence over the five generations since the death (in 1860) of the man who is believed to have had the first case of heart disease in the family, clearly indicates a dominant non-sex-linked mode of transmission.

The diagnosis of heart disease in each case was

based on the clinical findings and the electrocardiographic and radiologic changes. Two patients were asymptomatic and in these the only evidence for disease was electrocardiographic.

Dyspnoea on effort and syncopal attacks were the most common complaints. Sudden death occurred in eight instances. Frequent minor and major episodes of cerebral damage are considered to have resulted from emboli.

Significant abnormal physical findings were confined to the heart. Of the twenty patients with heart disease who were examined, rhythm was sinus in all but one. Nine had presystolic sounds, fifteen had systolic ejection murmurs, two had early diastolic murmurs, and six had mid-diastolic murmurs at the apex.

Electrocardiographic evidence of left ventricular hypertrophy or left bundle branch block was present in seven patients, but in most cases flat or diphasic T waves were the only abnormalities recorded. Cardiac enlargement was observed radiologically in fifteen patients and was predominantly left ventricular and left atrial. A frequent radiologic finding was an indentation seen in the left anterior oblique projection, identifying the junction of hypertrophied left ventricle and left atrium. The aortic shadow was considered to be abnormally small in eight patients.

Autopsies performed on three patients have revealed abnormalities restricted to the cardiovascular system. Hypertrophy and fibrosis of the myocardium, and hypoplasia of the aorta, have been constant findings. In addition, in two patients an unusual localized hypertrophy of the interventricular septum was found in which the hypertrophied myocardial fibres possessed a bizarre haphazard architectural arrangement. The cardiac valves and coronary arteries showed no disease.

The literature on idiopathic cardiomegaly and familial heart disease has been reviewed, and the various forms previously reported are compared with the disease observed in this family. In recent years an increasing number of reports on subaortic stenosis caused by asymmetrical hypertrophy of the interventricular septum indicate a growing awareness of this entity as a significant cardiac malformation. The observations on this family lend much support to previous contentions that the disease will frequently be familial, and it is hoped will stimulate further interest in the assessment of the relatives of subjects found to be affected.

Attempts have recently been made elsewhere to attack the hypertrophied septum surgically to relieve the subaortic stenosis. Insufficient time has elapsed to allow adequate assessment of the effectiveness of this procedure in prolonging life, but remedial surgery is being considered for the more severely affected members of the family under discussion, whose prognosis is, in any event, extremely poor.

The varied nomenclature that has been applied to this disease in the past, with both functional and organic connotations, has led to a confusion of terminology. The clinical and pathologic characteristics are now well delineated, but since the clinical findings are difficult to summarize in titular form, a descriptive term based on the pathologic anatomy appears desirable. "Hereditary cardiovascular dysplasia" is suggested as the most descriptive and least cumbersome name available.

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ADDENDUM

Since our paper was submitted for publication, many articles [44-53] on familial cardiopathy have appeared in the literature.

CASE HISTORIES

CASE 1. Jean B. (II). This man died suddenly in his thirties. He had three brothers and four sisters, all of whom lived to at least fifty years of age, and none of whom were known to have suffered from heart disease. His father lived to the age of fifty or sixty years and apparently was not seriously ill during his lifetime. Jean B. was married in 1851 and died sometime in the early 1860's. This information was obtained from a ninety-five year old nephew.

CASE 2. Jean B. (II). This man, the only child of Jean B. (Case 1), died at the age of thirty-six, with a known history of heart disease. The last three weeks of his life were spent sitting in a chair because of orthopnoea. He had two daughters and three sons. One daughter died at the age of sixty-four in a mental institution; there is no information as to the state of her

health or the cause of her death. The second daughter died at the age of seventy, and is known to have died with auricular fibrillation and a terminal episode of embolism to her lower extremities. One son died at the age of seventy from a cerebrovascular accident. The other two sons are (Cases 3 and 4) to be discussed. This information was obtained from his ninety-five year old cousin and from two daughters-in-law.

CASE 3. Jas. B. This man, son of Jean B. (Case 2), died suddenly at the age of thirty-seven while working on his farm. He had two children by each of two marriages, one child of each marriage died in infancy; cause of death was unknown. A son and daughter are still living. The case history on this man was obtained from his second wife.

CASE 4. Alf. B. This man, son of Jean B. (Case 2), died at the age of forty-six, two years after the onset of an illness described by his doctor as "dilated heart." For one and a half years before his death he was too disabled to work: he became more and more dyspneic before death and was unable to exert himself during the last year of his life. He never had swelling of the ankles and was never orthopneic. He became paralyzed on the day of his death. Of his thirteen children, one daughter has not been examined but is known to have heart disease and is forty-six years of age. Another daughter and son have been examined and were found to be free of disease. The other sons and daughters will be described as separate cases (Cases 5-12, 15 and 22). The information on Alf B. was obtained from his seventy-four year old wife who states that she had, in addition to her thirteen children, three miscarriages.

CASE 5. Marie B. This woman, daughter of Alf B. (Case 4), died at the age of forty-five, with a long history of heart disease. She had "dropsy" and required frequent paracenteses. She was very dyspneic and became paralyzed six months before death. She had two children, neither of whom have been examined. The information on this patient was obtained from her mother.

CASE 6. Alire B. This man, son of Alf B. (Case 4), died at the age of forty. Symptoms began at the age of thirty-two and consisted of severe cough and shortness of breath, which were recognized as being due to heart disease when he was thirty-six years of age. At that time he had a severe syncopal attack, and because of this and marked shortness of breath, he was advised to stop his work. During the last four years of his life he was an invalid, but did not show any orthopnoea or heart failure during most of this time. One year prior to death hemiparesis developed, and during the last six months of his life he was in heart failure. He died in 1949 but no autopsy was performed. However, a summary obtained from the

hospital in which he died described a diastolic murmur in the aortic area, and an electrocardiogram showed left axis deviation, prolonged P-R interval of 0.24 second, and QRS of 0.12 second. Of his ten children, one son has not yet been examined. Five daughters and two sons have been examined and have been found to be free from heart disease. One daughter and one son have been found to have heart disease, and will be discussed (Cases 19 and 20). The history of this man was obtained from his wife.

CASE 7. Eva B. (Mrs. V. B.) This woman, daughter of Alf B. (Case 4), died in 1943 at the age of thirty-three years. Disability dated from the age of eight years when she first complained of shortness of breath. During her married life she was always short of breath and required help to do her housework. She was never orthopneic and never manifested the signs of cardiac failure. Just prior to her death she suffered severe attacks of vomiting, and died suddenly following one of these attacks. She was not paralyzed. No autopsy was performed. Of her two children, one son was examined and found to be free from heart disease. Her daughter will be described (Case 14). In addition to the two living children, this woman had three miscarriages. The case history was obtained from her husband and her mother.

CASE 8. Gerard B. This man, son of Alf B. (Case 4) died in 1956 at the age of forty-three years. He worked as a delivery man for a bakery and was accustomed to climbing fifty sets of stairs each morning without disability until the Winter of 1956. At that time he became weak, a cough developed, and he had digestive upsets. In the Summer of 1956 a severe pain developed in the chest which lasted for three days and which was followed by haemoptysis. Following this episode he was confined to bed with cough, dyspnoea, orthopnoea and fatigue. On the day prior to his death a severe headache developed which was followed by a left-sided paralysis and aphasia. An electrocardiogram showed wide notched P waves (0.14 second) with sinus rhythm and right bundle branch block. Of three daughters, two were found to be normal on examination; the third will be described (Case 21). The history on this man was obtained from his wife.

CASE 9. Albert B. This man, son of Alf B. (Case 4), died at the age of thirty-five years, his history dating from the age of eighteen years at which time a temporary episode of aphasia and paralysis developed. He recovered completely from this episode and remained well for the next three years. At the age of twenty he was turned down for military service because of his heart. Dyspnoea was troublesome for six years prior to his death, but he was never orthopneic. He complained a great deal of digestive trouble with gaseous eructations, and had transitory attacks of weakness or "blackouts" which, however, did not

go on to syncope. During the last four months of his life these "blackouts" became so disabling that he had to give up his work. He died suddenly one day while working in his garden and was observed to have frothy expectoration terminally. Of his five children, two boys and one girl were found to be normal on examination. Two other boys have been found to have heart disease and will be described (Cases 29 and 30). The information on this man was obtained from his wife.

CASE 10. Therese B. This girl, daughter of Alf B. (Case 4), died suddenly at the age of twelve years on her way to church. She had complained of dyspnoea for about five to six months before death. On the morning of her death she appeared to be in good health when she left home, but asked her older sister to slow up because she was walking too fast. Suddenly she said, "I see everything black," and fell down and died immediately. There was no history of orthopnoea, syncope, cardiac failure or paralysis. Her case history was obtained from her mother and from the sister who was present at the time of her death.

CASE 11. Wilfred B. This man, son of Alf B. (Case 4), had no symptoms until the age of thirty-six when he noticed dyspnoea and some pressure in the chest while on a hunting trip. These symptoms did not increase and were not accompanied by orthopnoea or palpitations. He was admitted to a hospital three years later in September 1957, with complaints of numbness in the right side of the face and right arm. He showed no evidence of cardiac failure when first seen, but his heart was moderately enlarged on clinical examination. He had an atrial sound, a normal first sound, and a widely split second sound. There were no murmurs. The blood pressure was 120/80 mm. Hg. By radiologic examination all chambers of the heart were enlarged, predominantly the left ventricle. A prominent notch was observed at the left atrioventricular junction as seen in the left anterior oblique projection, compatible with hypertrophy of these two chambers. The aortic knob was small. There was evidence of pulmonary venous congestion. The electrocardiogram showed a normal sinus rhythm with occasional ventricular extrasystoles. The P waves were notched and slurred. The QRS complexes were those of an incomplete left bundle branch block. The urinalysis was normal. His haemoglobin was 100 per cent. The white blood count was 7,000 per cu. mm. with a normal differential. Roentgenograms of the skull, blood sugar, blood Wassermann test, serum non-protein nitrogen, electroencephalogram and cerebrospinal fluid examinations were all normal. Serum cholesterol was 110 mg. per cent (normal values 150 to 250 mg. per cent), total lipids 515 mg. per cent. The patient was given anticoagulants for the demonstrated cerebral vascular disease and was discharged to the care of his own doctor. Subsequently several cerebral

vascular accidents developed with right-sided hemiplegia and aphasia, and he was confined to bed at home where he died in May 1959. During the last few months of his life cardiac failure developed and he was treated with digitalis and diuretics. Permission for autopsy was obtained. This man had three sons and one daughter, of whom two sons were considered to have normal hearts. One son and the daughter will be discussed (Cases 27 and 28).

At autopsy the heart (680 gm.) was found to be enlarged, with dilatation of all chambers. The myocardium of the right ventricle was normal; that of the left was thickened, measuring 10 mm. at the apex and 14 mm. at its mid-portion. The interventricular septum was of normal width and did not contain a localized hypertrophied muscle mass. Multiple areas of fibrosis were seen scattered throughout the interventricular septum and left ventricle, particularly on their anterior aspects where the endocardium was thickened and a mural thrombus was firmly attached.

On microscopic examination the muscle fibres were hypertrophied and showed some disintegration with replacement by connective tissue. The architectural pattern did not appear to be unusual. The aorta was narrowed and was 2.4 cm. in diameter, 5.2 cm. in circumference, and 2 mm. in thickness above the aortic valve. At the arch of the aorta the measurements were, respectively, 2.3 cm., 4.8 cm. and 2 mm. There was slight thickening of the middle leaflet of the mitral valve, but no stenosis, and the other valves were all normal. The coronary arteries were patent throughout. There was hypoplasia of the muscle coat of the aorta and there was a moderate degree of arteriosclerosis. Evidence of old and recent infarcts were found in the lungs, the kidneys, the brain and the intestines.

CASE 12. *Rene B.* This man, son of Alf. B. (Case 4), had no symptoms except for vague epigastric discomfort until July 1957, when he had severe retrosternal pain which confined him to bed for a week. At the end of that time he felt well enough to return to work. About one month after the pain he had a severe grand mal seizure and left hemiplegia developed for which he was admitted to the hospital in August 1957, at the age of forty-one. On examination at that time, in addition to his left paresis, he was orthopneic and had Cheyne-Stokes breathing. His blood pressure was 98/50 mm. Hg and the heart showed sinus rhythm at the rate of 80. His heart was not reported as clinically enlarged but he had a widely split second sound at the base and an early diastolic murmur. Radiologically, cardiac enlargement was slight to moderate in degree and was predominantly left ventricular. The aorta was small. There was a considerable degree of pulmonary vascular congestion. The electrocardiogram showed sinus rhythm with ventricular extrasystoles, some of them in runs of bigeminy. The P wave was wide and

notched. The QRS complexes were abnormal, being widely splintered in the pattern of non-specific intraventricular block. The QRS was 0.12 second wide in lead aVL.

On admission, the urine showed albumin and pus cells which disappeared after treatment for cystitis. The haemoglobin was 86 per cent, and the white blood count, initially 21,000 per cu. mm. with neutrophilia, fell to 9,000 per cu. mm. at the time of discharge. Roentgenograms of the skull, blood sugar, blood Wassermann test and serum non-protein nitrogen were within normal limits. Serum cholesterol was 104 mg. per cent. During his hospital stay he recovered some strength in his left leg and was discharged home where he lived for one year. He was mildly dyspneic until a few days before his death when he became acutely dyspneic and had several attacks of pulmonary oedema. He died in December 1958, two and a half years after the onset of symptoms. No autopsy was performed. His only two children will be described (Cases 13 and 26).

CASE 13. *Claude B.* This boy, son of Rene B (Case 12), had symptoms of heart disease from infancy, when he would smother in his crying and become blue. He had shortness of breath all his life, which prevented him from engaging in any strenuous sports, although he was able to go to school. He never had orthopnoea, cough or palpitations. On two occasions he suddenly lost consciousness while exerting himself and was reported to have been unconscious for ten minutes. A doctor discovered a heart murmur when he was seven years old. On examination he was a well developed boy of eighteen, looking healthy. The abnormal findings were confined to the cardiovascular system. He had a forceful apical impulse which was bifid or reduplicated in character. The first sound was widely split. He had a grade 3/6 pansystolic murmur and a short, faint mid-diastolic murmur, heard best at the apex. The blood pressure was 110/40 mm. Hg. Radiologically, he had an increased ventricular mass, thought to represent both right and left ventricular hypertrophy, a moderately enlarged left atrium, and a prominent notch at the left atrioventricular groove. The main pulmonary artery was slightly enlarged but there was no pulmonary venous congestion. The aorta was small. The electrocardiogram showed tall and pointed P waves in leads II, III and aVF. The QRS complexes were of a high voltage in a pattern of left bundle branch block. He died suddenly in January 1959, at the age of eighteen.

Postmortem examination revealed a heart which was moderately enlarged and weighed 575 gm. There was myocardial hypertrophy involving all chambers but especially the left ventricle. The thickness of the left ventricle varied from 12 mm. at the apex to 20 mm. at the mid-portion. There was a localized nodular hypertrophy of the cephalic two-thirds of the inter-

ventricular septum which measured 32 mm. in thickness. This mass bulged into the right ventricle, distorting its outlines, narrowing the lumen and producing a globular shape to the heart. It also bulged, to a lesser extent, into the left ventricle, narrowing the outlet. The aorta looked small and narrow and measured 2 cm. in diameter and 2 mm. in thickness 1 cm. above the aortic valve. At the aortic arch the diameter was 2 cm. and the thickness 1 mm. The coronary arteries, the valves of the heart and the endocardium were all within normal limits.

On microscopic examination the myocardial fibres were hypertrophied and there was a peculiar haphazard architectural arrangement. In addition to this bizarre distribution of muscle fibres there was also some disintegration of the cytoplasm. Multiple and irregular areas of scarring were found scattered in the myocardium of both ventricles and interventricular septum. Microscopic examination also revealed the aorta and the medium-sized arteries to have thin walls with hypoplastic muscle coat containing some basophilic mucoid substance.

CASE 14. Noella B. This twenty year old girl, daughter of Eva B. (Case 7), died suddenly in church in February 1959. Her history was similar to that of her mother (Case 7). From the age of six years she was subjected to attacks of shortness of breath and weakness, accompanied by coughing and sweating, about once a month. A doctor who saw her then said that she had heart disease. The severe attacks disappeared at the age of eight years but she continued to have weakness and pain in the legs. Her dyspnoea was such that she could scarcely walk on the flat without becoming short of breath, but she never had orthopnoea, syncopal attacks or swollen ankles. On physical examination she was well developed, and abnormal findings were confined to the cardiovascular system. She had a widespread systolic murmur, 3/6. Her blood pressure was 120/76 mm. Hg. Roentgenographic examination showed enlargement of both ventricles and the left atrium. She had a prominent notch at the left atrioventricular junction. The aorta was small. The electrocardiogram showed diphasic P waves in leads V_1 and V_2 and abnormal notching of the QRS complexes, the QRS being 0.09 second wide. The T waves in leads V_6 and V_7 were flat.

A postmortem examination was performed on the embalmed body of this patient and the organs were in a rather poor state of preservation. The heart weighed 505 gm. There was mild hypertrophy of both atria, moderate hypertrophy of the right ventricle, which was 5 mm. thick, and marked hypertrophy of the left ventricle and interventricular septum in its cephalic two-thirds. The left ventricle measured 20 mm. in its mid-portion and 8 mm. at the apex. The interventricular septum was 30 mm. thick at the level of the nodular thickening. The localized hypertrophy of the

interventricular septum protruded into the lumens of both ventricles, narrowing their outlets and distorting the shape of the heart. The aorta appeared narrowed and was 2.4 cm. in diameter above the aortic valve and 2 mm. in thickness in this area. At the arch of the aorta the measurements were 2.6 cm. for the diameter and 2 mm. for the thickness of the aortic wall. The valves of the heart, although somewhat destroyed by the process of embalming, were considered to be normal. There was no evidence of any disease of the coronary arteries. The endocardium of all heart chambers was smooth and transparent. Microscopically, there was marked hypertrophy of the muscle fibres and a bizarre distribution of these fibres which seemed to lack organization. Disintegration of muscle fibres was also present. Multiple areas of scarring were noted throughout the myocardium. The muscle coats of the aorta and the medium-sized arteries were hypoplastic and showed an accumulation of slightly basophilic and mucoid substance between thinned muscle fibres.

CASE 15. Lauza (Mrs. L. G.) This fifty year old woman, daughter of Alf. B. (Case 4), had few symptoms and was able to perform all her housework without difficulty, but she believed she was weaker and had more dyspnoea than other women of her age. On examination she appeared to be healthy. Her blood pressure was 140/80 mm. Hg; her heart was in sinus rhythm at a rate of 84 per minute. Her heart sounds were normal with the addition of a low-pitched atrial gallop. Roentgenographic examination of the chest was normal. The electrocardiogram was abnormal, showing slightly depressed S-T segments from leads V_4 to V_6 with diphasic T waves. Of her four children, two boys were believed to be free of heart disease; one boy and a girl had abnormal hearts and will be described (Cases 16 and 17).

CASE 16. Andre G. This boy, son of Lauza (Case 15), had no symptoms. On examination his blood pressure was 125/80 mm. Hg. He had a grade 1 systolic murmur. A roentgenogram of the chest was normal. An electrocardiogram showed right bundle branch block pattern.

CASE 17. Jeannine G. This eighteen year old girl, daughter of Lauza (Case 15), was found to have a heart murmur as an infant, and had symptoms of heart disease. As a child she had fainting attacks, fatigue, palpitations, dyspnoea and cough on the slightest exertion. On physical examination her heart was enlarged and fibrillating. She had a loud split first sound at the apex with a pansystolic murmur. After exercise she had an opening snap and a mid-diastolic murmur. Her blood pressure was 110/70 mm. Hg. She had an atrial fibrillation. On radiologic examination all cardiac chambers were considered enlarged, and there was considerable pulmonary

vascular congestion. The electrocardiogram showed auricular fibrillation with abnormal notching of the QRS complexes which were 0.09 second wide, and negative T waves in leads aVF and V_5 to V_7 . A tracing made just before death showed complete atrioventricular block. Death occurred following a brief period of heart failure. Autopsy was not performed.

CASE 18. Lise B. This eighteen year old girl, daughter of Joseph B. (Case 22), had no symptoms except frequent nosebleeds and a story of her legs becoming red and blue after standing for any length of time. The physical examination was within normal limits. Her blood pressure was 120/80 mm. Hg. A roentgenogram of the chest was normal. The electrocardiogram showed only a non-specific intraventricular conduction defect and a Q in lead aVF 9 mm. with an R of 23 mm.

CASE 19. Lucille B. This twelve year old girl, daughter of Alire B. (Case 6), was active and had no symptoms. An atrial and a third heart sound were heard. Radiologically, there was slight enlargement of both ventricles and of the left atrium. The electrocardiogram was abnormal in that it showed inverted T waves in leads aVF and V_5 to V_7 .

CASE 20. Henri B. This eleven year old boy, son of Alire B. (Case 6), had no symptoms until the age of ten years, when he noticed unusual dyspnoea and weakness on taking part in active sports. On examination he had a grade 3/6 rough mid-systolic murmur heard over the entire precordium, and a third heart sound which was elicited by exercise. The blood pressure was 130/85 mm. Hg. Radiologic examination revealed slight enlargement of both right and left ventricles. A left atrioventricular notch was present posteriorly, and the size of the aorta was at the lower limits of normal. The electrocardiogram showed S-T depression in leads V_2 to V_6 and negative T waves in leads I, aVL and V_5 to V_7 .

CASE 21. Jacinthe B. This twelve year old girl, daughter of Gerald B. (Case 8), had shortness of breath and weakness on slight exertion since the age of six years, at times severe enough to keep her in bed. She had pain in her legs, particularly in bed at night. On examination she was pale, but well developed and well nourished. She had an atrial sound, a high-pitched mid-systolic murmur grade 2/6, and a mid-diastolic murmur heard at the apex. Her blood pressure was 105/65 mm. Hg. Radiologically, there was enlargement of all cardiac chambers, graded as slight to moderate in degree. The aorta was small. A second roentgenogram taken a year later showed an increase in cardiac size during that interval. An electrocardiogram was abnormal, showing biphasic T waves in leads II, III and aVF and inverted T waves in leads V_4 to V_7 . The serum cholesterol level was 112 mg. per cent.

CASE 22. Joseph B. This forty-four year old man, son of Alf. B. (Case 4), had no symptoms other than complaints of poor digestion and frequent nosebleeds. On examination he was well developed; a faint atrial sound and a systolic murmur grade 2/6 were heard. The blood pressure was 135/85 mm. Hg. A roentgenogram of the chest was within normal limits. The electrocardiogram was abnormal, showing negative T waves in leads I, II and V_5 to V_7 . His serum cholesterol level was 212 mg. per cent. Of his six children, one girl and one boy were believed to be free of heart disease. Three girls and one boy will be discussed (Cases 18, 23, 24 and 25).

CASE 23. Marcelle B. This seventeen year old girl, daughter of Joseph B. (Case 22), had dyspnoea on exertion. On physical examination she was a plump, well developed girl. Her heart was not clinically enlarged, but she had an apical third sound heard after exercise. Her blood pressure was 124/80 mm. Hg. Radiologic examination showed slight enlargement of the right and left ventricles and of the left atrium, with an indentation marking the junction of the left atrium and the left ventricle. The electrocardiogram was abnormal, with a QS pattern in leads V_2 and V_3 and negative T waves in leads II, aVF and V_5 to V_7 . The serum cholesterol level was 171 mg. per cent.

CASE 24. Gisele B. This fifteen year old girl, daughter of Joseph B. (Case 22), had no symptoms and was active in sports. On examination she had only a faint mid-systolic murmur, heard best at the apex, and a third heart sound heard after exercise. The blood pressure was 120/75 mm. Hg. Radiologically, there was a slight to moderate enlargement of both ventricles and of the left atrium, with left atrioventricular notching posteriorly. The electrocardiogram was abnormal, showing a negative T in leads II, III, aVF and V_4 to V_6 . The serum cholesterol level was 204 mg. per cent.

CASE 25. Yves B. This eleven year old boy, son of Joseph B. (Case 22), had had weakness and dyspnoea on exertion since the age of two years. He was thin and slightly pigeon breasted. He had a grade 2/6 apical systolic murmur and a loud third heart sound heard at the apex. His blood pressure was 120/70 mm. Hg. Roentgenographic examination of the chest showed slight but definite enlargement of the left atrium and possible enlargement of the left and right ventricles. The electrocardiogram was within normal limits. The serum cholesterol level was 141 mg. per cent.

CASE 26. Leo B. This sixteen year old boy, son of Rene B. (Case 12), had been quite well until one year previously when he first noted shortness of breath while playing hockey. At about the same time he had an acute febrile illness and a syncopal attack. He did not complain of orthopnoea or weakness. On examina-

tion he was a muscular young man. He had a systolic thrill, an atrial sound, a 4/6 mid-systolic murmur and a short mid-diastolic murmur. Radiological examination performed in 1958 and repeated in 1959 showed enlargement of both ventricles and atria, with a slight increase in the heart size during the year. He had prominent notching at the junction of the left atrium and ventricle. The aorta was small. The electrocardiogram was abnormal, showing diphasic P waves in leads V_3R to V_3 , high voltage QRS complexes, abnormal Q waves in leads aVF and V_6 to V_7 . He was admitted to the Royal Victoria Hospital for further investigation in April 1959. Urinalysis, protein electrophoresis, serum electrolytes, blood sugar, cardiac output at rest and with exercise, and blood volume were all found to be within normal limits. His white blood count was 13,400 per cu. mm. with slight leukocytosis and eosinophilia. He had slightly reduced prothrombin activity. His blood group was O positive. Skin tests for toxoplasmosis, histoplasmosis and tuberculosis were negative. A tomogram of the lungs revealed normal vasculature. Pulmonary function tests showed normal ventilation, diffusion, arterial blood oxygen and carbon dioxide. A roentgenogram of the skull and an electroencephalogram were within normal limits. The serum cholesterol level was 153 mg. per cent.

CASE 27. Denis B. This fourteen year old boy, son of Wilfred B. (Case 11), was asymptomatic until the age of ten years when he had an isolated syncopal attack. At about thirteen years he started noticing shortness of breath on exertion, first while playing actively, later on with only moderate activity. He had no orthopnoea. On examination he had a grade 1/6 systolic murmur. Roentgenographic examination of the chest revealed no abnormalities. The electrocardiogram was abnormal, showing an inverted T wave in leads aVF and V_4 to V_6 .

CASE 28. Claire B. This eighteen year old girl, daughter of Wilfred B. (Case 11), had only vague symptoms of dyspnoea when first seen in 1958, but by 1959 not only had dyspnoea developed while walking on the level but also orthopnea and oedema of the ankles. She had a dry cough. On physical examination she was well developed. There was a mid-systolic murmur along the left sternal border and a third heart sound heard after exercise. Roentgenographic examination of the chest was within normal limits in 1958 but in 1959 showed slight but definite enlargement of all cardiac chambers. The electrocardiogram was abnormal, with inverted T waves in leads aVF and V_5 to V_7 .

CASE 29. Luc B. This sixteen year old boy, son of Albert B. (Case 9), was found to have a heart murmur at the age of three years when he was admitted to the hospital for investigation of repeated epistaxes. He

had never been strong and suffered continuously from a dry cough, shortness of breath on exertion, and blackouts or giddiness on unusual straining. At the age of thirteen years he became orthopneic after a severe infection of the lower respiratory tract and was admitted to another hospital for investigation. He was reported to have had an elevated pulmonary artery and capillary pressure on cardiac catheterization. On physical examination he was muscular and well developed. He had a 3/6 mid-systolic murmur at the apex and a third heart sound accompanied by a short mid-diastolic murmur at the apex. The blood pressure was 120/60 mm. Hg. Radiological examination in 1958 showed only a slight enlargement of all chambers, with an indentation between the left atrium and ventricle. By 1959 further enlargement had occurred, particularly of the ventricles. The aorta was smaller than normal. His electrocardiogram was abnormal, with tall or notched P waves (4 mm. in lead II), a Q wave in leads aVL 0.04 seconds wide and 8 mm. tall (R 14 mm.). T waves were negative in leads I, aVL and V_5 to V_7 . A more complete investigation was carried out at the Royal Victoria Hospital and showed a low cardiac output of 3.3 L./min. at rest, with an increase of only 0.6 L. on exercise. Vital capacity and functional residual capacity were slightly decreased. Pressure-volume studies revealed normal lung mechanics. Arterial oxygen saturation and carbon dioxide were normal. An electroencephalogram showed a paroxysmal dysrhythmia seen maximally in the temporal posterior and central head regions. Roentgenograms of the skull were within normal limits. Prothrombin activity was slightly decreased. The serum cholesterol level was 96 mg. per cent, well below the normal range of 150 to 250 mg. per cent. White blood count, differential, haemoglobin and red cells were all within the normal range, as were protein electrophoresis, serum electrolytes, blood sugar, urinalysis and blood volume.

CASE 30. Roch B. This fourteen year old boy, son of Albert B. (Case 9), was said by his mother to have been short of breath since the age of six or seven. At the time we saw him he was unable to play hockey for long without becoming acutely dyspneic. He had had one transitory syncopal attack, but suffered repeated attacks of dizzy spells on exertion. On examination he was pale but of stocky build. He had a bifid apical impulse and an apical systolic thrill. The heart sounds were normal, with an atrial sound, a 4/6 systolic murmur heard maximally along the left sternal border, and a third heart sound accompanied by a short mid-diastolic murmur at the apex. The electrocardiogram was abnormal, showing high voltages of the QRS complexes, the R wave 50 mm. and the S wave 60 mm. in lead V_3R . He had S-T depression in leads I, aVL, V_4 to V_7 and negative T waves in the same leads. Roentgenographic examination of the chest in 1958 showed slight enlargement of all cardiac chambers, which had

increased on a repeat examination in 1959. A left atrioventricular notch was present posteriorly, and the aortic size was small. He was admitted to the Royal Victoria Hospital for further investigation. Serum protein, electrophoresis, electrolytes, blood sugar, Wassermann test, urinalysis, electroencephalogram, skin tests for toxoplasmosis, histoplasmosis and tuberculosis, tomography of the lungs and roentgenograms of the skull were all normal. He had a normal cardiac output and blood volume. Prothrombin activity was slightly decreased. The serum cholesterol level was 106 mg. per cent. Results of pulmonary function tests were normal, except for a slight decrease in the vital capacity and the functional residual capacity.

REFERENCES

- NADAS, A. S., ALIMURUNG, M. M. and SIERACKI, L. A. Cardiac manifestations of Friedreich's ataxia. *New England J. Med.*, 244: 239, 1951.
- EVANS, W. and WRIGHT, G. Electrocardiogram in Friedreich's disease. *Brit. Heart J.*, 4: 91, 1942.
- NOTHACKER, W. G. and NETSKY, M. G. Myocardial lesions in progressive muscular dystrophy. *Arch. Path.*, 50: 578, 1950.
- BERENBAUM, A. A. and HOROWITZ, W. Heart involvement in progressive muscular dystrophy. *Am. Heart J.*, 51: 622, 1956.
- GAILANI, S., DANOWSKI, T. S. and FISHER, D. S. Muscular dystrophy: catheterization studies indicating latent congestive heart failure. *Circulation*, 17: 583, 1958.
- LISAN, P., IMBRIGLIA, J. and LIKOFF, W. Myocardial disease associated with progressive muscular dystrophy (a report of two cases). *Am. Heart J.*, 57: 913, 1959.
- LEVY, R. L. and VON GLALIN, W. C. Cardiac hypertrophy of unknown cause; a study of the clinical and pathologic features in ten adults. *Am. Heart J.*, 28: 714, 1944.
- LEVIN, E. B. and COHEN, S. L. Idiopathic myocardial hypertrophy simulating rheumatic heart disease. *Am. Heart J.*, 48: 637, 1954.
- DAVIES, R. K., MARVEL, B. J. and GENOVESE, P. D. Heart disease of unknown etiology. *Am. Heart J.*, 42: 546, 1951.
- ELSTER, S. K., HORN, H. and TUCHMAN, L. R. Cardiac hypertrophy and insufficiency of unknown etiology. *Am. J. Med.*, 18: 900, 1955.
- SERBIN, R. A. and CHOJNACKI, B. Idiopathic cardiac hypertrophy. *New England J. Med.*, 252: 10, 1955.
- FRERE ELOI-GERARD. *Genealogies des Familles de Beauce, Dorchester et Frontenac*, vol. 1, Beauceville, Quebec, 1949. L'Eclaireur.
- ADDARIO, F., MARTINI, L., MAHAIM, I. and WINSTON, M. Nouvelles recherches sur le "bloc bilatéral manqué." *Cardiologia*, 11: 36, 1946-1947.
- EVANS, W. Familial cardiomegaly. *Brit. Heart J.*, 11: 68, 1949.
- DAVIES, L. G. A familial heart disease. *Brit. Heart J.*, 14: 206, 1952.
- GAUNT, R. T. and LECUTIER, M. A. Familial cardiomegaly. *Brit. Heart J.*, 18: 251, 1956.
- CAMPBELL, M. and TURNER-WARWICK, M. Two more families with cardiomegaly. *Brit. Heart J.*, 18: 393, 1956.
- PAULLEY, J. W., JONES, R., GREEN, W. F. D. and KANE, E. P. Myocardial toxoplasmosis. *Brit. Heart J.*, 18: 55, 1956.
- SOMMERS, B. Problems in the clinical diagnosis and classification of ventricular hypertrophy in adults. III. Familial cardiomyopathy. *Minnesota Med.*, 39: 153, 1956.
- NIKKILA, E. A. and PELKONEN, R. Isolated (Fiedler's) myocarditis and idiopathic cardiac hypertrophy. *Acta med. scandinav.*, 165: 421, 1959.
- SHAH, P. M. Familial cardiomegaly: case reports with review of literature. *Indian J. Med. Sc.*, 11: 818, 1957.
- SOULIE, P., DIMATTEO, J., ABABA, A., NOUAILLE, J. and THIBERT, M. H. Cardiomegalie familiale. *Arch. mal coeur*, 50: 22, 1957.
- KREMER, R. and DELCHEVALERIE, J. Cardiomegalie familiale avec blocs de branche. *Cardiologica*, 12: 527, 1957.
- FACQUET, J., ALHOMME, P., WELTI, J.-J., BONNET, J.-L. and TINTURIER, J. Cardiomegalie familiale apparemment benigne affectant plusieurs enfants d'une nombreuse famille. *Bull. et mém. Soc. méd. d. hôp. de Paris*, 74: 80, 1958.
- SOULIE, P., DIMATTEO, J. and NOUAILLE, J. Cardiomegalie familiale. *Bull. et mém. Soc. méd. d. hôp. de Paris*, 73: 1091, 1957.
- FAIVRE, G., DEBRY, G., GILGENKANTZ, J.-M., PANTHU, J.-J. and SCHWARTZ, G. Cardiomegalie familiale. *Rev. méd. Nancy*, 83: 1075-1093, 1958.
- BRIGDEN, W. Uncommon myocardial diseases—the non-coronary cardiomyopathies. *Lancet*, 273: 1179, 1243, 1957.
- TEARE, D. Asymmetrical hypertrophy of the heart in young adults. *Brit. Heart J.*, 20: 1, 1958.
- BERCU, B. A., DIETERT, G. A., DANFORTH, W. H., PUND, E. E., AHLVIN, R. C. and BELLIVEAU, R. Pseudoaortic stenosis produced by ventricular hypertrophy. *Am. J. Med.*, 25: 814, 1958.
- BRENT, L. B., FISHER, D. L. and TAYLOR, W. J. Familial muscular sub-aortic stenosis. (Abstract.) *Circulation*, 20: 676, 1959.
- BRACHFELD, N. and GORLIN, R. Subaortic stenosis: a revised concept of the disease. *Medicine*, 38: 415, 1959; 32 & 33 (old 21 & 22).
- BRAUNWALD, E., MORROW, A. C., CORNELL, W. F., AVGEN, M. M. and KILBISH, T. F. Idiopathic hypertrophic subaortic stenosis: clinical, hemodynamic and angiographic manifestations. *Am. J. Med.*, 29: 924, 1960.
- MORROW, A. G. and BRAUNWALD, E. Functional aortic stenosis: a malformation characterized by resistance to left ventricular outflow without anatomic obstruction. *Circulation*, 20: 181, 1959.
- GOODWIN, J. F., HOLLMAN, A., CLELAND, W. P. and TEARE, D. Obstructive cardiomyopathy simulating aortic stenosis. *Brit. Heart J.*, 22: 403, 1960.
- BROCK, R. C. Functional obstruction of the left ventricle. *Guy's Hosp. Rep.*, 106: 221, 1957.
- GREENBERG, J. and SIMON, M. A. Subaortic stenosis in an adult. *Canad. Med. A. J.*, 61: 50, 1949.
- AVERILL, J. H., RANDALL, R. V., JOFFE, E. and WHITE, P. D. Heart disease of unknown cause in siblings. *Ann. Int. Med.*, 47: 49, 1957.

38. THOMAS, W. A., RANDALL, R. V., BLAND, E. F. and CASTLEMAN, B. Endocardial fibroelastosis: a factor in heart disease of obscure etiology—a study of 20 autopsied cases in children and adults. *New England J. Med.*, 251: 327, 1954.
39. BLUMBERG, R. W. and LYON, R. A. Endocardial sclerosis. *Am. J. Dis. Child.*, 84: 291, 1952.
40. VAN BUCHEM, F. S. P. Arachnodactyly heart. *Circulation*, 20: 88, 1959.
41. WHITFIELD, A. G. W., ARNOTT, W. M. and STAFFORD, J. L. "Myocarditis" and aortic hypoplasia in arachnodactyly. *Lancet*, 1: 1387, 1951.
42. TASCHEN, B. Familiäre Herztotenfälle in Jugendalter. *Arch. path. Anat.*, 323: 39, 1953.
43. HAYES, W. L., DIMOND, E. G. and CROCKETT, J. E. Idiopathic cardiomegaly with extreme cardiac enlargement—hypoplastic aorta primary or secondary factor? *Am. J. Cardiol.*, 4: 528: 1959.
44. VORIDES, E. and MAURICE, P. Cardiomegalie familiale. *Arch. mal. coeur*, 52: 1410, 1959.
45. SCHIEBLER, G. L. ADAMS, P. and ANDERSON, R. C. Familial cardiomegaly in association with Wolff-Parkinson-White syndrome. *Am. Heart J.*, 58: 113, 1959.
46. GARRETT, G., HAY, W. J. and RICKARDS, A. G. Familial cardiomegaly. *J. Clin. Path.*, 12: 355, 1959.
47. WALTHER, R. J., MADOFF, I. M. and ZINNER, K. Cardiomegaly of unknown cause occurring in a family. Report of three siblings and review of the literature. *New England J. Med.*, 263: 1104, 1960.
48. BACOS, J. M., EAGAN, J. T. and ORGAIN, E. S. Congenital familial nodal rhythm. *Circulation*, 22: 719, 1960.
49. BATTERSBY, E. J. and GLENNER, G. G. Familial cardiomyopathy. *Circulation*, 22: 721, 1960.
50. BEASLEY, O. L. Familial myocardial disease. A report of three siblings and a review of the literature. *Am. J. Med.*, 29: 476, 1960.
51. HOLLMAN, A., GOODWIN, J. F., TEARE, D. and RENWICK, J. A family with obstructive cardiomyopathy (asymmetrical hypertrophy). *Brit. Heart J.*, 22: 449, 1960.
52. FOWLER, N. O., GUERON, M. and ROWLANDS, D. T., JR. Idiopathic cardiomyopathy. *Circulation*, 22: 751, 1960.
53. WHITFIELD, A. G. W. Familial cardiomyopathy. *Quart. J. Med.*, 30: 119, 1960.